Guideline

Guidelines for File Assessment for Pharmaceutical Products for Human Use.

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1. Introduction:

The role of the CADC is integral to the oversight of the national regulatory authority, and its contribution encompasses an integrated chain of activities, throughout the lifecycle of a medical product. By virtue of its responsibility for the laboratory testing function in the Egyptian Drug Authority, CADC's contribution is evident in the performance of various activities, including the evaluation and assessment of the quality part of product dossiers submitted to CADC, to provide technical and scientific input before marketing authorization is granted for a product, renewal of MA or re-registration, and post-approval variations.

The EDA aims to strengthen its regulatory system and align itself with regulatory authorities recognized by the WHO, to achieve harmonization with their regulatory practices, with the ultimate objective of ensuring the safety, efficacy and quality of those products. To this end, and in alignment with EDA's strategic objectives, this guideline has been elaborated to clarify the technical requirements for the assessment of product analytical files, based on criteria adopted from ICH guidelines, FDA guidelines, WHO guidelines, as well as international pharmacopoeias.

2. Scope

The guidelines detailed in this document apply to product files of finished pharmaceutical products for human use; submitted by the manufacturer/ MAH to CADC for laboratory testing. It applies to both locally produced and imported, which have been granted marketing authorization (registration/re-registration license), or are still in the process of receiving a registration/re-registration license, for analytical file renewal and for post-approval changes.

3. Abbreviations

3.1 BP: British Pharmacopoeia

3.2 CADC: Central Administration of Drug Control

3.3 CAPP: Central Administration of Pharmaceutical Products

3.4 EA: Administration of Evaluation and Approval

3.5 EDA: Egyptian Drug Authority

3.6 Egy.Ph: Egyptian Pharmacopoeia

3.7 EMA: European Medicine Agency

3.8 FDA: Food & Drug Administration

3.9 FPP: Finished Pharmaceutical Product

3.10 Ph. Int.: International Pharmacopoeia

3.11 JP: Japanese Pharmacopoeia

3.12 MA: Marketing authorization

3.13 MAH: Marketing Authorization Holder

3.14 NCL: National Control Laboratory

3.15 NRA: National Regulatory Authority

3.16 PAC: Administration of Post Approval Control

3.17 PDE: Permissible Daily Exposure

3.18 Ph. Eur.: European Pharmacopoeia

3.19 TAE: Administration of Technical Assessment and Evaluation

3.20 USP: United States Pharmacopoeia

3.21 WLA: WHO-Listed Authority

4. Definitions

4.1 Finished Pharmaceutical Product: A finished dosage form of a pharmaceutical product is known to be the product that has undergone all production stages, including packaging in its final container and labeling.

4.2 Final report: a certificate of analysis of a pharmaceutical product that is issued from CADC, and includes the product specifications that have been approved for the marketing authorization of the product. The Final Report is attached to the product registration file archived in CADC.

4.3 Pharmacopeial product: A product that has a specific monograph set forth in an official pharmacopeia (such as the USP, BP, or EP). The trade name includes an abbreviation indicating which pharmacopeia it adheres to.

5. Main Topic

5.1 The manufacturer/ MAH is required to upload the requisite documents using the link specified on EDA's website upon application for laboratory testing.

5.2 Document review and technical assessment shall be performed by a delegated team of qualified reviewers.

5.3 The pharmacopoeias that are recognized for CADC are Egy.Ph, USP, BP, Ph.Eur, Ph.Int and JP.

5.4 For the purpose of organizing assessment activities, product analytical files that are submitted to CADC for technical assessment are categorized on a procedural and technical basis into two groups:

5.4.1 Group I

Products which are either locally produced or imported, that have previously received a registration license submitted to the Administration of Post Approval Control laboratory testing for purposes other than post approval changes/variations.

5.4.2 Group II

Products which are either locally produced or imported, submitted as part of proceedings for registration, re-registration, post approval variations or analytical file renewal, submitted to Administration of Evaluation and Approval, or products submitted to the Administration of Post Approval Control, for post approval variations.

5.5 Group I general rules:

5.5.1 Document review shall be carried out with reference to the Final Report previously issued by the EA for registration of the product, (or the updated specifications approved by TAE for approval of MA renewal, or the detailed final report issued by the PAC for post approval changes, or the approved quality module specifications issued by the central administration of pharmaceutical products or any other approved specifications as a part of the registration license, whichever is pertinent to the case).

5.5.1.1 In case the final report is not available, review will be according to product specifications approved by the General Administration for Stability in the CAPP.

5.5.1.2 In case there are no acceptance limits for one or more of the tests specified in the Final Report previously issued from CADC, the manufacturer/MAH is required to add the test limits to the product specifications according to the specifications approved by the General Administration for Stability in the CAPP or according to the pharmacopoeia limits (Egy.Ph, USP, BP, Ph.Eur, Ph.Int or JP), with no stipulation for the manufacturer to apply to CADC for modification of the previously issued Final Report.

5.5.1.3 The following tests need to be added, or if present, the acceptance limits need to be updated, if applicable to the dosage form:

- Dissolution rate test: acceptance limits shall be as listed in Annex I and detailed in Annex II.
- Bacterial endotoxin: acceptance limits shall be according to pharmacopeial limits (Egy.Ph, USP, BP, Ph.Eur, Ph.Int or JP), or according to Annex IV (Requirements for Microbiological Analysis, section 5).
- Particulate matter: acceptance limits shall be according to pharmacopeial limits (Egy.Ph, USP, BP, Ph.Eur, Ph.Int or JP).

5.5.1.4 If the manufacturer/MAH wishes to amend (delete - add - change limits) for one of the tests, they are directed to the Administration of Variation in CAPP and the relevant rules and regulations must be applied.

5.5.1.5 When MA is renewed for a product, with updated product or package specifications, and where laboratory testing is not stipulated by the Administration of Variation or the Variation Committee for approval, the previously issued Final Report stands, and the updated specifications shall be attached to the product registration file archived in CADC for future reference.

5.5.2 Laboratory testing is performed according to the analytical methods that have been previously approved in CADC for MA, and that are attached to the product registration file archived in CADC. In case there are changes in the analytical method/s, the applicant is required to declare such change and upload the modified method accompanied with complete validation data (or verification, if pharmacopeial) and payment receipt via the link specified on the website for method update.

5.5.3 Files for imported or locally produced FPPs may be submitted by the applicant via the link for pre-submission assessment.

5.6 Group II general rules:

5.6.1 Approvals and decisions issued by any of the scientific and technical committees of EDA shall be taken into consideration in the decision-making process in CADC

5.6.2 Whenever a pharmacopoeia is used as a reference, this shall always refer to the most recent version thereof at the time of manufacturing of the finished product.

5.6.3 If there is a monograph for the finished product, and if the monograph specifies certain tests that are not stated in this guideline, the manufacturer/MAH should add those tests, or otherwise justify waiving those tests.

5.6.4 Water loss test which is evaluated for the aqueous-based products packaged in semi-permeable containers as a stability test will not be included in the final report.

5.6.5 Complete comparative dissolution study in different media may be required.

5.6.6 In the case of imported FPPs for human use that are approved by one or more of the countries listed in the Technical Committee for Drug Control's list of reference countries:

5.6.6.1 The products may be considered for the reliance pathway according to the regulatory guideline issued for the EDA Chairman's Decree No 780 for the Year 2020 (at the discretion of the applicant), whereby the products are assessed and analyzed according to their specifications that have been previously approved by the reference country's NRA with acceptable scientific justification

5.6.6.2 The applicant will submit the required documents via the link for pre-submission assessment.

5.6.7 If the FPP is a pharmacopeial product, the manufacturer/MAH shall adhere to the tests and acceptance criteria stated in the product's monograph, in addition to those in guidelines.

5.6.8 The products are assessed and analyzed according to shelf-life specifications.

5.7 General Rules

5.7.1 Composition

5.7.1.1 The reference specified in the product composition (BP, USP etc.) must comply with the registration license.

5.7.1.2 The function of inactive materials in product should be clarified according to Handbook of Pharmaceutical Excipients or any other reliable reference.

5.7.1.3 For a pharmacopeial API, it should comply with the latest version of the specified pharmacopoeia.

5.7.2 Calculation sheet:

5.7.2.1 There should be a separate calculation sheet to calculate equivalency of salt to the base.

5.7.2.2 For substances for which the potency is calculated as international units, the amount of the substance will be stated in the product composition in international units and denoted

with (*) and it should be clarified in the footnote below the table that the amount used depends on the potency of the raw material.

5.7.3 Registration form

5.7.3.1 A full description of the package, concordant with the samples received for analysis, should be stated.

5.7.3.2 The name of the manufacturer should be stated.

5.7.4 The finished product specification and certificate of analysis of production should contain the active material as stated in registration license and product composition.

5.7.5 Method of analysis

In case of using updated method, the following shall be submitted:

5.7.5.1 Full detailed method.

5.7.5.2 Complete validation or verification protocol and report.

5.7.5.3 Complete validation or verification charts.

5.7.5.4 Receipt of fees payment to change the method.

5.7.6 Whenever a manufacturer chooses not to perform a test included in this guideline, a scientific justification should be submitted.

6 References:

6.1 Egyptian Pharmacopoeia, 5th edition, 2025 (Egy.Ph).

6.2 United States Pharmacopeial Convention Committee of Revision, 2025, USP-NF Online (48th edition.).

6.3 British Pharmacopoeia Commission. British Pharmacopoeia 2025, Online.

6.4 International Pharmacopoeia, 11th edition, 2025.

6.5 European pharmacopoeia, EP 11th edition.

6.6 Japanese pharmacopoeia, JP 18th edition.

6.7 ICH Q6A - Specifications: Test procedures and acceptance criteria for new drug substances and new drug products: Chemical substances.

6.8 ICH Q2 Validation of Analytical Procedures.

6.9 ICH Q3A Impurities in New Drug substances.

6.10 ICH Q3B Impurities in New Drug Products.

6.11 ICH Q3C Guidance for Residual Solvents Guidance for Industry

6.12 OMCL (Validation of Analytical Procedures PA/PH/OMCL (13) 82 2R).

6.13 Food and Drug Administration, "Methods, Method Verification and Validation", Laboratory Manual, ORA Laboratory Procedure, Volume II, ORA-LAB.5.4.5.

6.14 FDA Guidance for industry: Dissolution Testing and Acceptance Criteria for Immediate-Release Solid Oral Dosage Form Drug Products Containing High Solubility Drug Substances. AUGUST 2018.

6.15 FDA guidance for industry: Dissolution Testing of Immediate Release Solid Oral dosage form. AUGUST 1997.

6.16 FDA guidance for industry: Nasal Spray and Inhalation Solution, Suspension, and Spray Drug Products — Chemistry, Manufacturing, and Controls Documentation Center for Drug Evaluation and Research (CDER). July 2002.

6.17 WHO annex 6 Guidelines on submission of documentation for a multisource (generic) finished pharmaceutical product: quality part.

7 Annexes:

7.1 Annex I: Requirements for Physical Analysis

7.2 Annex II: Development of the In-house Dissolution Methods and Comparative Dissolution Study

7.3 Annex III: Requirements for Chemical Analysis

7.4 Annex IV: Requirements for Microbiological Analysis

7.5 Annex V: Method Validation and Verification

7.6 Annex VI: Required Documents for File Submission to CADC

Annex I

Requirements for Physical Analysis

File assessment for physical analysis of any dosage form will be performed according to the following tables:

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1. Tests performed on Aerosols (packaged under pressure):

Test	Applicability	Acceptance criteria
1. Description	All	According to manufacturer specifications
2. Minimum fill (Egy.Ph, USP) Procedure according to: Egy.Ph (B.3.31) USP-NF ⟨755⟩	All	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3. Leak rate (Egy.Ph, USP, BP) Procedure according to: Egy.Ph (B.3.46) USP-NF ⟨604⟩ Ph. Eur. Monograph 0671 Ph. Eur. Monograph 0676	○ Topical aerosols fitted with continuous valves (Egy.Ph, USP) ○ Metered dose inhalation and nasal aerosols (Egy.Ph, USP, BP)	Leak Rate Egy.Ph (B.3.46) USP-NF ⟨604⟩ Ph. Eur. Monograph 0671 Ph. Eur. Monograph 0676
4. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	Inhalation and nasal aerosols	According to manufacturer specifications
5. Valve delivery (shot weight test) (Egy.Ph, USP) Procedure according to: Egy.Ph (B.6.5) Inhalation and Nasal Drug Products—General Information and Product Quality Tests USP-NF ⟨5⟩	Perform these tests only on inhalation and nasal aerosol (metered dose)	According to manufacturer specifications
6. No. of deliveries per container (Egy.Ph, USP, BP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩ Ph. Eur. Monograph 0671 Ph. Eur. Monograph 0676	Perform this test for aerosols fitted with dose-metering valves	The total number of deliveries discharged is not less than the number stated on the label.
7. Delivery rate (Egy.Ph, USP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩	Continuous valve topical aerosols	According to manufacturer specifications
8. Delivered amount (Egy.Ph, USP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩	Continuous valve topical aerosols	According to manufacturer specifications

Test	Applicability	Acceptance criteria
9. Droplet/Particle size Distribution by laser diffraction (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests Egy.Ph (B.6.16) USP-NF ⟨601⟩	○ Nasal aerosol suspension (particle size) ○ Nasal aerosol solution (droplet size)	According to manufacturer specifications
10. Spray pattern/ Plume geometry (Egy.Ph, USP) (Shape and size of evolving spray) Procedure according to Inhalation and Nasal Drug Products—General Information and Product Quality Tests Egy.Ph (B.6.5) USP-NF ⟨5⟩	○ Nasal and inhalation aerosol	According to manufacturer specifications
11. Pressure test (pressure gauge) (Egy.Ph, USP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩	○ Continuous valve topical aerosols	According to manufacturer specifications
12. Aerodynamic particle size measurement (cascade impactor) (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests. Egy.Ph (B.6.16) USP-NF ⟨601⟩	○ Inhalation aerosol	According to manufacturer specifications

2. Tests performed on capsules:

Test	Applicability	Required Information	Acceptance criteria
1. Description: ○ **Appearance** ○ **Colour**	All	○ Capsule type: hard gelatin capsule/soft gelatin capsule ○ Capsule size ○ Colour of cap: acc. to supplier. ○ Colour of body: acc. to supplier. ○ Type and colour of content (powder, pellets, liquid, granules, paste…etc).	According to manufacturer specifications
2. Mass uniformity* (BP) Procedure according to: Ph. Eur. method 2.9.5	Done on capsule content only. If the test for uniformity of content is prescribed for all the active substances, the test for uniformity of mass is not required**.**		● Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation. ● None deviate by more than twice that percentage. (Ph. Eur. method 2.9.5)
3. Disintegration time (Egy.Ph, USP, BP) Procedure according to Disintegration: Egy.Ph (B.3.29) USP-NF ⟨701⟩ Ph. Eur. method 2.9.1 Ph. Eur. method 2.9.2	All except capsules for inhalation		Disintegration Egy.Ph (B.3.29) USP-NF ⟨701⟩ (Ph. Eur. monograph 0016) (Ph. Eur. monograph 1164)
4. Dissolution** Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph*** ○ FDA dissolution methods database. (submit comparative dissolution profile in the most suitable media)**** ○ In-house method Obey the dissolution method development criteria**** Refer to (Annex II)	For all ○ Disintegration could substitute dissolution as a performance test if a justification submitted by the manufacturer that it obeys the ICH Q6A guidelines. In this case, the performed dissolution method should be supplied by the manufacturer.	● Dissolution Parameters: ○ Media composition, pH & volume ○ Apparatus type ○ RPM ○ Temperature ○ Sinkers needed (common type: coiled sinker) ○ Filter type (common types: Nylon, PVDF & PTFE) ○ Sampling time(s) ○ Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3

Test	Applicability	Required Information	Acceptance criteria
5. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Needs justification to skip test		According to manufacturer specifications
6. Acid-neutralizing capacity (Egy.Ph, USP) Procedure according to Acid neutralizing capacity: Egy.Ph (B.4.1) USP-NF ⟨ 301⟩	Antacids only		According to manufacturer specifications

* Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.
** In case of locally acting API (not systemically absorbed), dissolution rate test may not be done and disintegration time is sufficient.
*** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).
**** Reflection paper on the dissolution specification for generic solid oral immediate release products with systemic action EMA/CHMP/CVMP/QWP/336031/2017 may be referred to as guidance.

3. Tests performed for creams, gels & ointments:

Test	Applicability	Required Information	Acceptance criteria
1.Description: ○ **Appearance** ○ **Colour** ○ **Homogeneity** ○ **Visible foreign matter**	All		According to manufacturer specifications
2.Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	For single and multiple dose units N.B. In case of single unit containers where the test for content uniformity is applied, the test for minimum fill is not required.		Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	○ O/W cream ○ Aqueous gel ○ Hydrophilic ointment Generally: it is Formulation dependent. According to manufacturer specifications Because some topically applied drug products contain very limited quantities of water or aqueous phase, pH measurements may not always be warranted.	● Kind of product ○ Hydrophilic or ○ Lipophilic ● Preparation method to perform measurement: ○ Solvent ○ Percent of dilution	According to manufacturer specifications
4. Apparent viscosity (Egy.Ph, USP) Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	All	○ Type of device (model) ○ Device subtype ○ Spindle no. ○ RPM ○ Temperature	According to manufacturer specifications
5.Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test		According to manufacturer specifications

Test	Applicability	Required Information	Acceptance criteria
6. Particle size (BP) Procedure according to: (Ph.Eur.1163) using microscope.	Semi-solid ophthalmic preparations contain dispersed solid particles.		• Not more than 20 particles have a maximum dimension greater than 25 µm, and not more than 2 of these particles have a maximum dimension greater than 50 µm. • None of the particles has a maximum dimension greater than 90 µm. (Ph.Eur.1163)

4. Tests performed on emulsions:

Test	Applicability	Acceptance criteria
1.Description: ○ **Appearance** ○ **Colour** ○ **Viscous or not**	All	According to manufacturer specifications
2.Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	○ Vaginal emulsion ○ Rectal emulsion ○ Ophthalmic emulsion ○ Otic emulsion ○ Topical emulsion	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.Deliverable volume (Egy.Ph, USP) Procedure according to Deliverable Volume: Egy.Ph (B.3.30) USP-NF ⟨698⟩	Oral emulsions (labeled volume should be known) If the test of uniformity of dosage units is prescribed for single unit containers, the test for deliverable volume is not required	Deliverable Volume Egy.Ph (B.3.30) USP-NF ⟨698⟩
4.pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to: Manufacturer's method.	Hydrophilic emulsions (o/w) It is formulation dependent, according to manufacturer specifications.	According to manufacturer specifications
5.Specific gravity/viscosity (Egy.Ph, USP) Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	Relatively viscous emulsions ○ Ophthalmic emulsion ○ Topical emulsion o Otic emulsion o Oral emulsion ○ Vaginal emulsion ○ Rectal emulsion	According to manufacturer specifications
6.Uniformity and accuracy of delivered doses from multidose containers Procedure according to: Ph. Eur. Method 2.9.27	Oral emulsions which are supplied in multi-dose containers provided at manufacture with a measuring device.	● Not more than 2 of the individual masses or the individual volumes deviate from the average mass or the average volume by more than 10 % and none deviates by more than 20 %. (Ph. Eur. method 2.9.27)
7.Uniformity of dose of oral drops (BP) Procedure according to: Liquid Preparations for Oral Use, Ph. Eur. monograph 0672.	Oral drops only	Liquid Preparations for Oral Use, Ph. Eur. monograph 0672

Test	Applicability	Acceptance criteria
8.Container content for injection (Egy.Ph, USP)/ Extractable volume (BP) Procedure according to Container Content for Injections: Egy.Ph (B.3.66) USP-NF ⟨697⟩ Ph. Eur. Method 2.9.17	Parenteral emulsion	Container Content for Injections Egy.Ph (B.3.66) USP-NF ⟨697⟩ Extractable Volume of Parenteral Preparations Ph. Eur. Method 2.9.17
9. Globule size	○ Ophthalmic emulsion ○ Parenteral emulsion	According to manufacturer specifications
10.Osmolality (Egy.Ph, USP, BP) Procedure according to Osmolality and Osmolarity: Egy.Ph (B.3.58) USP-NF ⟨785⟩ (Ph. Eur. method 2.2.35)	○ Ophthalmic emulsion ○ Parenteral emulsion*	According to manufacturer specifications
11. Container–closure integrity	Parenteral emulsions	Package Integrity Leak Test Technologies ⟨1207.2⟩, Package Seal Quality Test Technologies ⟨1207.3⟩

* When the tonicity of a product is declared in its labeling and/or when composition includes tonicity. It is also done for large volume parenteral.

5. Tests performed on films:

Test	Applicability	Required Information	Acceptance criteria
1.Description: ○ **Appearance** ○ **Dimensions**	All		According to manufacturer specifications
2. Mass uniformity* Procedure according to: Ph. Eur. method 2.9.5	If the test for uniformity of content is prescribed for all the active substances, the test for uniformity of mass is not required.		● Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation. ● None deviate by more than twice that percentage. (Ph. Eur. method 2.9.5)
4.Dissolution Reference of method is one of the following: ○ Egy.Ph, USP or BP specific monograph** ○ FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). ○ In-house method (submit comparative dissolution profile in the most suitable media).	All	• Dissolution Parameters: ○ Media composition, pH & volume ○ Apparatus type ○ RPM ○ Temperature ○ Filter type (common types: Nylon, PVDF & PTFE) ○ Sampling time(s) ○ Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3
5. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) Procedure according to manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip the test.		According to manufacturer specifications.

* Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.
** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).

6. Tests performed on foams:

Tests	Applicability	Acceptance criteria
1.Description **Physical appearance (of the foam and of the collapsed foam) (USP)**	All	According to manufacturer specifications
2. Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	All	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.Leak rate (Egy.Ph, USP) Procedure according to: Egy.Ph (B.3.46) USP-NF ⟨604⟩	All	Egy.Ph (B.3.46) Leak Rate USP-NF ⟨604⟩
4.pH (Egy.Ph, USP, BP) Egy.Ph (B.3.23) pH USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to: Manufacturer's method.	For the collapsed foam It is a formulation dependent, according to manufacture specifications	According to manufacturer's specifications
5.Relative density of uncollapsed Foam (USP) Procedure according to Pharmaceutical Foams Product Quality Tests: USP-NF ⟨607⟩	Topical	According to manufacturer's specifications
6.Time to Break (USP) Procedure according to Pharmaceutical Foams Product Quality Tests: USP-NF ⟨607⟩	Topical	According to manufacturer's specifications
7.Delivery rate (Egy.Ph, USP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩	Topical	According to manufacturer's specifications
8.Delivered amount (Egy.Ph, USP) Procedure according to Topical Aerosols: Egy.Ph (B.3.67) USP-NF ⟨603⟩	Topical	According to manufacturer's specifications
9. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to Manufacturer's method or specific monograph.	Primarily for nonaqueous formulations If not stated by manufacturer: Need justification to skip the test	According to manufacturer's specifications

Tests	Applicability	Acceptance criteria
10. Osmolality (USP) Procedure according to USP-NF ⟨785⟩ Osmolality and Osmolarity	If applicable	According to manufacturer's specifications
11. Pressure test (USP) Procedure according to: USP-NF ⟨607⟩ Pharmaceutical Foams Product Quality Tests.	All	According to manufacturer's specifications

7. Tests performed on granules:

Test	Applicability		Required Information	Acceptance criteria
	Granule Type	Done/ Not Done		
1. Description: ○ **Appearance** ○ **Colour** ○ **Visual clarity (for solution of granules after reconstitution).**	All		○ Colour of granules before and after reconstitution ○ Solution or suspension after reconstitution (with certain viscosity or not)	According to manufacturer specifications
2. Deliverable volume (Egy.Ph, USP) Procedure according to Deliverable Volume: Egy.Ph (B.3.30) USP-NF ⟨698⟩	Only <u>oral granules</u> for reconstitution (after reconstitution) in: ○ Multiple dose container ○ Single dose container <u>Not done for granules that are administered with food or beverages only</u>. If the test of uniformity of dosage units is prescribed for single unit containers, the test for deliverable volume is not required	○ Yes ○ Yes	Labeled volume	Deliverable Volume Egy.Ph (B.3.30) USP-NF ⟨698⟩
3. Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	○ Granules for oral suspension packaged in containers (where test of deliverable volume is applicable). ○ Other multiple dose granules.	○ No ○ Yes	Labeled amount	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
4. Mass uniformity* (BP) Procedure according to: Ph. Eur. method 2.9.5	○ Uncoated single dose granules ○ Coated granules ○ Multiple dose granules If the test for uniformity of content is prescribed or justified and authorized for all the active substances, the test for uniformity of mass is not required.	○ Yes ○ No ○ No		• Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation. • None deviates by more than twice that percentage. (Ph. Eur. method 2.9.5)
5. Uniformity and accuracy of delivered doses from multidose containers Procedure according to: Ph. Eur. Method 2.9.27	○ Oral granules which are supplied in multi-dose containers provided at manufacture with a measuring device.			• Not more than 2 of the individual masses or the individual volumes deviate from the average mass or the average volume by more than 10 % and none deviates by more than 20 %. • (Ph. Eur. method 2.9.27)

Test	Applicability		Required Information	Acceptance criteria
	Granule Type	Done/ Not Done		
6. Dissolution** Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph*** ○ FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). In-house method (submit comparative dissolution profile in the most suitable media).	○ Granules that result in an oral suspension.		• Dissolution Parameters: ○ Media composition, pH & volume ○ Apparatus type ○ RPM ○ Temperature ○ Filter type (common types: Nylon, PVDF & PTFE) ○ Sampling time(s) Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3 •
7. Disintegration time (Egy.Ph, USP, BP) Procedure according to Disintegration: Egy.Ph (B.3.29) USP-NF ⟨701⟩ (Ph. Eur. monograph 0499)	○ Effervescent granules			Disintegration Egy.Ph (B.3.29) USP-NF ⟨701⟩ • (Ph. Eur. monograph 0499)
8. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test ○			• According to manufacturer specifications
9. pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	○ For reconstituted granules (after reconstitution). Except granules that are administered with food or beverages only.			• According to manufacturer specifications
10.Redispersibility/Resuspendability Procedure according to: ICH Q6A USP-NF ⟨2⟩	○ For granules reconstituted to form suspensions not intended for immediate use			• According to manufacturer specifications

Test	Applicability		Required Information	Acceptance criteria
	Granule Type	Done/ Not Done		
11. Uniformity of dose of oral drops (BP) Procedure according to: Liquid Preparations for Oral Use, Ph. Eur. monograph 0672.	○ For granules intended to give oral drops after reconstitution.			Liquid Preparations for Oral Use, • Ph. Eur. monograph 0672
12. Specific gravity/Viscosity (Egy.Ph, USP) Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	For relatively viscous reconstituted suspensions (after reconstitution) ○ Ophthalmic suspension ○ Otic suspension ○ Topical suspension ○ Oral suspension ○ Nasal suspension ○ Vaginal suspension ○ Rectal suspension			• According to manufacturer specifications
13. Acid neutralizing capacity (Egy.Ph, USP) Procedure according to Acid neutralizing capacity: Egy.Ph (B.4.1) USP-NF ⟨ 301⟩	○ For antacids			• According to manufacturer specifications

* Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.
** In case of locally acting API (not systemically absorbed), dissolution rate test may not be done.
*** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).

8. Tests performed on lozenges:

Test	Applicability		Information should be available	Acceptance criteria
	Lozenge type	Done/ Not Done		
1.Description: ○ **Appearance** ○ **Colour** ○ **Molded or compressed**	All		○ Lozenge shape ○ Color ○ Biconvex/flat	According to manufacturer specifications
2.Mass uniformity*	○ Molded ○ Compressed	○ Yes ○ Yes		According to manufacturer specifications
3.Water content (Egy.Ph, USP) Water determination USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test			According to monograph or manufacturer's specifications
4. Dissolution Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph** ○ FDA dissolution methods database FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). ○ In-house method (submit comparative dissolution profile in the most suitable media).	○ Molded ○ Compressed for local effect ○ Compressed for systemic effect	○ No ○ No ○ Yes	• <u>Dissolution Parameters:</u> • Media composition, pH & volume • Apparatus type • RPM • Temperature • Filter type (common types: Nylon, PVDF & PTFE) • Sampling time(s) • Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3
5.Friability (USP & BP) Procedure according to Tablet Friability: USP-NF ⟨1216⟩ Ph. Eur. method 2.9.7	○ Molded ○ Compressed	○ No ○ Yes		Tablet Friability USP-NF ⟨1216⟩ Ph. Eur. method 2.9.7
6.Hardness (USP& BP) Procedure according to Tablet Breaking Force: USP-NF ⟨1217⟩ Ph. Eur. method 2.9.8	○ Molded ○ Compressed	○ No ○ Yes		According to manufacturer's specifications

*Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.

** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).

9. Tests performed on powders:

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
1.Description: ○ **Appearance** ○ **Colour** ○ **Visual clarity (for solution of powder after reconstitution).**	All		○ Colour of powders before and after reconstitution ○ Solution or suspension after reconstitution with certain viscosity or not	According to manufacturer specifications
2.Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	○ Powders for oral suspension packaged in containers (where test of deliverable volume is applicable). ○ Other multiple dose powders . ○ Powder for inhalation (device metered)	○ No ○ Yes ○ Yes	Labeled amount	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.Deliverable volume (Egy.Ph, USP) Procedure according to Deliverable Volume: Egy.Ph (B.3.30) USP-NF ⟨698⟩	Only oral powders for reconstitution (after reconstitution) in: ○ Multiple dose container ○ Single dose container If the test of uniformity of dosage units is prescribed for single unit containers, the test for deliverable volume is not required	○ Yes ○ Yes	Labeled volume	Deliverable Volume Egy.Ph (B.3.30) USP-NF ⟨698⟩

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
4. Uniformity and accuracy of delivered doses from multidose containers Procedure according to: (Ph.Eur. method 2.9.27)	Oral powders which are supplied in multi-dose containers provided at manufacture with a measuring device. (Done for all doses)			Not more than 2 of the individual masses or the individual volumes deviate from the average mass or the average volume by more than 10 % and none deviates by more than 20 %. (Ph. Eur. method 2.9.27)
5. Mass uniformity* (BP) Procedure according to: (Ph. Eur. method 2.9.5).	○ Single dose powders ○ Powders for parenteral administration (single dose) ○ Powders for eye-drops and powders for eye lotions (single-dose) ○ If powder for parenteral administration average mass ≤40 mg If the test for uniformity of content is prescribed or justified and authorized for all the active substances, the test for uniformity of mass is not required.	○ Yes ○ Yes ○ Yes No		● Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation ● None deviates by more than twice that percentage. (Ph. Eur. method 2.9.5)
6. Disintegration time (BP) Procedure according to: (Ph. Eur. Monograph 1165)	Effervescent powders			(Ph. Eur. monograph 1165)
7. Dissolution** Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph*** ○ FDA dissolution methods database (submit comparative dissolution profile in the most	○ Powder reconstituted to form oral suspension. ○ Powder reconstituted to form sustained	○ Yes ○ Yes	•<u>Dissolution Parameters:</u> • Media composition, pH & volume • Apparatus type • RPM • Temperature • Filter type (common types: Nylon, PVDF & PTFE) • Sampling time(s)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
suitable media). In-house method (submit comparative dissolution profile in the most suitable media).	ophthalmic or parenteral suspension.		Q (the amount dissolved)	
8. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test			According to manufacturer specifications
9. Reconstitution time (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products—General Information and Product Quality Tests Egy.Ph (B.6.5) USP-NF ⟨5⟩ ICH Q6A	○ Powder for inhalation solution Powder for parenteral			According to manufacturer specifications
10. pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	For reconstituted powders (after reconstitution).			According to manufacturer specifications
11. Particulate matter (Egy.Ph, USP) Procedure according to Particulate Matter in Injections: Egy.Ph (B.3.43) USP- NF ⟨788⟩ Particulate Matter in Ophthalmic Solutions Egy.Ph (B.3.44) USP-NF ⟨789⟩	Powder and lyophilized powders for parenteral solutions and intra/extra ocular injections.			Particulate Matter in Injections for Extra-ocular solutions for injections and for parenteral solutions Egy.Ph (B.3.43) USP-NF ⟨788⟩ Particulate Matter in Ophthalmic Solutions for intra-ocular solutions for injections Egy.Ph (B.3.44) USP-NF ⟨789⟩

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
12. Completeness of solution after reconstitution (Egy.Ph, USP) Procedure according to Completeness of solution: Egy.Ph (B.3.22) USP-NF ⟨641⟩ Inhalation and Nasal Drug Products— General Information and Product Quality Tests: Egy.Ph (B.6.5) USP-NF ⟨5⟩ Injections and Implanted Drug Products (Parenteral)-Product Quality Tests: Egy.Ph (B.6.1) USP-NF ⟨1⟩	Powder for inhalation solution Powder for parenteral solution			Completeness of solution Egy.Ph (B.3.22) USP-NF ⟨641⟩ Inhalation And Nasal Drug Products— General Information And Product Quality Tests. Egy.Ph (B.6.5) USP-NF ⟨5⟩ Injections And Implanted Drug Products (Parenteral)-Product Quality Tests Egy.Ph (B.6.1) USP-NF ⟨1⟩
13. Redispersibility/Resuspendability Procedure according to: ICH Q6A USP-NF ⟨2⟩	For powders reconstituted to form suspensions not intended for immediate use			According to manufacturer specifications
14. Powder fineness (Egy.Ph, USP, BP) Procedure according to Powder fineness: Egy.Ph (B.3.33) USP-NF ⟨811⟩ Sieve test Ph. Eur.2.9.35	Topical powder	Done if prescribed (stated in the monograph or by manufacturer)		Powder fineness Egy.Ph (B.3.33) USP-NF ⟨811⟩ Ph. Eur.2.9.35
15. Uniformity of dose of oral drops (BP) Procedure according to: Liquid Preparations for Oral Use, Ph. Eur. monograph 0672	For powders intended to give oral drops after reconstitution.			Liquid Preparations for Oral Use, Ph. Eur. monograph 0672
16- Specific gravity/Viscosity (Egy.Ph, USP) Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩,	For reconstituted powder (after reconstitution) ○ Ophthalmic ○ Nasal ○ Inhalation ○ Topical ○ Otic and Oral			According to manufacturer specifications

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩				
17-Acid-neutralizing capacity (Egy.Ph, USP) Procedure according to Acid neutralizing capacity: Egy.Ph (B.4.1) USP-NF ⟨ 301⟩	For antacids			According To Manufacturer Specifications.
18- Particle size distribution** (Egy.Ph, USP)** Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	Powder for reconstitution to give parenteral suspension			According To Manufacturer Specifications.
19. Aerodynamic size distribution (cascade impactor, Marple Miller Impactor) (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	Inhalation powder			According to manufacturer specifications
20. Number of deliveries per container/per inhaler Procedure according to: Ph. Eur. monograph 0676 Ph. Eur. monograph 0671	Nasal powder supplied in multidose containers Inhalation powder supplied in multidose containers			The total number of deliveries discharged is not less than the number stated on the label.

Test	Applicability		Required Information	Acceptance criteria
	Powder Type	Done/ Not Done		
21. Plume Geometry (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products— General Information and Product Quality Tests: Egy.Ph (B.6.5) USP-NF ⟨5⟩	Nasal powder	If device is pump-dependent		According to manufacturer specifications

* Average weight could be considered if needed as IPC (USP-NF (1163) Quality assurance in pharmaceutical compounding).
** In case of locally acting API (not systemically absorbed), dissolution rate test may not be done.
*** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).
**** Particle size distribution testing may be proposed in place of dissolution testing, when development studies demonstrate that particle size is the primary factor influencing dissolution; justification should be provided. The acceptance criteria should include acceptable particle size distribution in terms of the percent of total particles in given size ranges. The mean, upper, and / or lower particle size limits should be well defined.

10. Tests performed on solutions:

Test	Applicability	Acceptance criteria
1.Description: ○ **Appearance** ○ **Colour** ○ **Visual clarity** ○ **Viscous or not.**	All	According to manufacturer specifications
2.Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	○ Nasal solution ○ Inhalation solution ○ Vaginal solution ○ Rectal solution ○ Ophthalmic solution ○ Otic solution ○ Topical solution	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.Mass uniformity (BP) Procedure according to Ph. Eur. Monograph 0671	Single-dose inhalation solutions	Ph. Eur. monograph 0671
4.pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	Aqueous solutions: it is formulation dependent, according to manufacturer specifications.	According to the manufacturer specifications
5. Specific gravity/Viscosity (Egy.Ph, USP) Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	Relatively viscous solutions ○ Ophthalmic solution ○ Nasal solution ○ Inhalation solution ○ Parenteral solution ○ Topical solution ○ Otic solution ○ Oral solution ○ Vaginal solution ○ Rectal solution	According to manufacturer specifications
6.Particulate matter (Egy.Ph, USP) Procedure according to Particulate Matter in Injections: Egy.Ph (B.3.43) USP- NF ⟨788⟩ Particulate Matter in Ophthalmic Solutions Particulate Matter in Ophthalmic Solutions Egy.Ph (B.3.44) USP-NF ⟨789⟩	○ Extra and intraocular solutions for injections ○ Parenteral solutions	Particulate Matter in Injections for Extra-ocular solutions for injections and for parenteral solutions Egy.Ph (B.3.43) USP-NF ⟨788⟩ Particulate Matter in Ophthalmic Solutions for intra-ocular solutions for injections Egy.Ph (B.3.44) USP-NF ⟨789⟩

Test	Applicability	Acceptance criteria
7. Uniformity and accuracy of delivered doses from multidose containers Procedure according to Ph. Eur. method 2.9.27	Oral solutions which are supplied in multi-dose containers provided at manufacture with a measuring device. ○ (Done for all doses)	Not more than 2 of the individual masses or the individual volumes deviate from the average mass or the average volume by more than 10 % and none deviates by more than 20 %. (Ph. Eur. method 2.9.27)
8. Uniformity of dose of oral drops (BP) Procedure according to Liquid Preparations for Oral Use, Ph. Eur. monograph 0672	○ Oral drops only	Liquid Preparations for Oral Use, Ph. Eur. monograph 0672
9. Deliverable volume (Egy.Ph, USP) Procedure according to Deliverable Volume: Egy.Ph (B.3.30) USP-NF ⟨698⟩	Oral solutions ○ If the test of uniformity of dosage units is prescribed for single unit containers, the test for deliverable volume is not required	Deliverable Volume Egy.Ph (B.3.30) USP-NF ⟨698⟩
10. Container content for injection (Egy.Ph, USP)/ Extractable volume (BP). Procedure according to Container Content for Injections: Egy.Ph (B.3.66) USP-NF ⟨697⟩	○ Parenteral solution	Container Content for Injections Egy.Ph (B.3.66) USP-NF ⟨697⟩
11. Osmolality (Egy.Ph, USP, BP) Procedure according to Osmolality and Osmolarity: Egy.Ph (B.3.58) USP-NF ⟨785⟩ (Ph. Eur. method 2.2.35)	○ Ophthalmic solutions ○ Inhalation solutions ○ Parenteral solutions* ○ Nasal solutions ○	According to manufacturer specifications
12. Container–closure integrity (USP)	○ Parenteral solutions	Package Integrity Leak Test Technologies ⟨1207.2⟩, Package Seal Quality Test Technologies ⟨1207.3⟩

* When the tonicity of a product is declared in its labeling and/or when composition includes tonicity. It is also done for large volume parenteral.

11. Tests performed on sprays (non-pressurized liquid):

Test	Applicability	Acceptance criteria
1. Description	All	According to manufacturer specifications
2. Uniformity of delivered mass (BP) Procedure according to Ph. Eur. monograph 0676 Ph. Eur. monograph 1807	○ Metered-dose nasal sprays and single dose containers that are solutions. ○ Metered dose oro-mucosal sprays and sublingual sprays that are solutions. If the test of uniformity of delivered dose or uniformity of content is prescribed or justified and authorized for all the active substances, the test for uniformity of delivered mass is not required.	Ph. Eur. monograph 0676 Ph. Eur. monograph 1807
3.(USP) **(Egy.Ph, USP)** Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	All	Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
4. Pump delivery (shot weight test) **(Egy.Ph, USP)/FDA** Procedure according to Inhalation and Nasal Drug Products—General Information and Product Quality Tests: Egy.Ph (B.6.5) USP-NF ⟨5⟩	Nasal and inhalation sprays (metered dose) If the test of uniformity of delivered mass and/or uniformity of delivered dose or uniformity of content is prescribed or justified and authorized for all the active substances, the test of pump delivery is not required	According to FDA*
5.pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	Formulation dependent, according to manufacturer specifications	According to manufacturer specifications
6. Specific gravity / Viscosity **(Egy.Ph, USP)** Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	For Nasal spray	According to manufacturer specifications

Test	Applicability	Acceptance criteria
7. Droplet/Particle size distribution by laser diffraction. (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	○ Nasal spray suspension (particle size) Nasal spray solution (droplet size)	According to manufacturer specifications
8. Aerodynamic particle size measurement (cascade impactor) (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	Inhalation spray only	According to manufacturer specifications
9.Osmolality (Egy.Ph, USP, BP) Procedure according to Osmolality and Osmolarity: Egy.Ph (B.3.58) USP-NF ⟨785⟩ (Ph. Eur. method 2.2.35)	Nasal spray	According to manufacture specifications
10. Number of deliveries per container/per inhaler Procedure according to: Ph. Eur. monograph 0676 Ph. Eur. monograph 0671	Nasal metered-dose preparations Inhalation metered-dose preparations	The total number of deliveries discharged is not less than the number stated on the label.
11. Spray pattern (Egy.Ph, USP) Inhalation and Nasal Drug Products—General Information and Product Quality Tests. (shape and size of evolving spray) Egy.Ph (B.6.5) USP-NF ⟨5⟩	Nasal spray	According to manufacture specifications
12. Plume geometry (Egy.Ph, USP) Inhalation and Nasal Drug Products—General Information and Product Quality Tests. (shape and size of evolving spray) Egy.Ph (B.6.5) USP-NF ⟨5⟩	Inhalation spray	According to manufacture specifications

*FDA guidance for industry: Nasal Spray and Inhalation Solution, Suspension, and Spray Drug Products — Chemistry, Manufacturing, and Controls Documentation Center for Drug Evaluation and Research (CDER). July 2002.

12. Tests performed on suppositories:

Test	Applicability		Required Information	Acceptance criteria
	Type	Done/ Not Done		
1. Description: ○ **Appearance** ○ **Colour**	All			According to manufacturer Specifications
2. Mass uniformity* (BP) Procedure according to Ph. Eur. method 2.9.5	All suppositories and pessaries If the test for uniformity of content is prescribed or justified and authorized for all the active substances, the test for uniformity of mass is not required.			● Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation. ● None deviates by more than twice that percentage. (Ph. Eur. method 2.9.5)
3. Disintegration time (Egy.Ph, BP) Procedure according to: Egy.Ph (B.3.29) Ph. Eur. method 2.9.1	Done for all unless intended for modified release or prolonged local action.			Egy.Ph (B.3.29) Ph. Eur. method 2.9.1
4. Dissolution ** Reference of method is one of the following: ○ Egy.Ph, USP or BP specific monograph*** ○ FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). ○ In-house method (submit comparative dissolution profile in the most suitable media).	All Suppositories and pessaries.		•<u>Dissolution Parameters:</u> • Media composition, pH & volume • Apparatus type • RPM • Temperature • Filter type (common types: Nylon, PVDF & PTFE) • Sampling time(s) Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3
5. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test			According to monograph or manufacturer specifications
6. Softening time (USP, BP, Ph. Int.)	Lipophilic rectal suppositories As an alternative test for the disintegration time test			According to monograph or manufacturer specifications

* Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.
** In case of locally acting API (not systemically absorbed), dissolution rate test may not be done.
*** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).

13. Tests performed on suspensions:

Test	Applicability	Required Information	Acceptance criteria
1. Description: ○ **Appearance** ○ **Homogeneity** ○ **Color/ with certain viscosity or not**	All		According to manufacturer Specifications
2. Minimum fill (Egy.Ph, USP) Procedure according to Minimum Fill: Egy.Ph (B.3.31) USP-NF ⟨755⟩	○ Nasal suspension ○ Inhalation suspension, ○ Vaginal suspension, ○ Rectal suspension, ○ Ophthalmic suspension, ○ Otic suspension. ○ Topical suspension.		Minimum Fill Egy.Ph (B.3.31) USP-NF ⟨755⟩
3.pH (Egy.Ph, USP, BP) pH Egy.Ph (B.3.23) USP-NF ⟨791⟩ Ph.Eur. (2.2.3) Procedure according to Manufacturer's method.	Aqueous suspensions It is formulation dependent, according to manufacturer specifications.		According to manufacturer specifications
4. Specific gravity/Viscosity (Egy.Ph, USP) Procedure according to specific gravity: Egy.Ph (B.3.26) USP-NF ⟨841⟩ Procedure according to manufacturer's method: Viscosity–Capillary Methods Egy.Ph (B.3.34) USP NF ⟨911⟩, Viscosity–Rotational Methods Egy.Ph (B.3.35) USP-NF ⟨912⟩, and Viscosity–Rolling Ball Method Egy.Ph (B.3.36) USP-NF⟨913⟩	Relatively viscous suspensions ○ Nasal suspension ○ Inhalation suspension ○ Ophthalmic suspension ○ Topical suspension ○ Otic suspension ○ Oral suspension ○ Vaginal suspension ○ Rectal suspension		According to manufacturer specifications
5. Uniformity and accuracy of delivered doses from multidose containers Procedure according to (Ph. Eur. method 2.9.27)	Oral suspensions which are supplied in multi-dose containers provided at manufacture with a measuring device. (Done for all doses)		Not more than 2 of the individual masses or the individual volumes deviate from the average mass or the average volume by more than 10 % and none deviates by more than 20 %. (Ph. Eur. method 2.9.27)
6.Uniformity of dose of oral drops (BP) Procedure according to (Liquid Preparations for Oral Use, Ph. Eur. monograph 0672)	Oral drops only		(Liquid Preparations for Oral Use, Ph. Eur. monograph 0672)
7. Deliverable volume (Egy.Ph, USP) Procedure according to Deliverable Volume: Egy.Ph (B.3.30)	Oral suspensions If the test of uniformity of dosage units is prescribed for single unit containers,		Deliverable Volume Egy.Ph (B.3.30) USP-NF ⟨698⟩

Test	Applicability	Required Information	Acceptance criteria
USP-NF ⟨698⟩	the test for deliverable volume is not required		
8. Container content (Egy.Ph, USP)/ Extractable volume (BP) Procedure according to Container Content for Injections: Egy.Ph (B.3.66) USP-NF ⟨697⟩	Parenteral suspension		Container Content for Injections Egy.Ph (B.3.66) USP-NF ⟨697⟩
9. Dissolution* Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph** ○ FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). In-house method (submit comparative dissolution profile in the most suitable media).	○ Oral suspensions (unless otherwise justified). ○ Sustained ophthalmic suspensions ○ Sustained parenteral suspensions	● Dissolution Parameters: • Media composition, pH & volume • Apparatus type • RPM • Temperature • Filter type (common types: Nylon, PVDF & PTFE) • Sampling time(s) Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3
10. Acid Neutralizing capacity (Egy.Ph, USP) Procedure according to Acid neutralizing capacity: Egy.Ph (B.4.1) USP-NF ⟨ 301⟩	Antacids		According to manufacturer specifications
11.Redispersibility/Resuspendability Procedure according to: ICH Q6A USP-NF ⟨2⟩	All suspensions		According to manufacturer specifications
12. Particle size distribution * (Egy.Ph, USP)** Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	○ Nasal suspension ○ Ophthalmic suspension ○ Parenteral suspension Inhalation suspension		According to manufacturer specifications
13. Aerodynamic particle size measurement (cascade impactor) (Egy.Ph, USP) Procedure according to Inhalation and Nasal Drug Products_ Aerosols, Sprays, and Powders—Performance Quality Tests: Egy.Ph (B.6.16) USP-NF ⟨601⟩	Inhalation suspension		According to manufacturer specifications

Test	Applicability	Required Information	Acceptance criteria
14.Osmolality (Egy.Ph, USP, BP) Procedure according to Osmolality and Osmolarity: Egy.Ph (B.3.58) USP-NF ⟨785⟩ (Ph. Eur. method 2.2.35)	○ Ophthalmic suspensions ○ Nasal suspensions ○ Inhalation suspensions ○ Parenteral suspensions****		According to manufacturer specifications
15. Container–closure integrity	Parenteral suspensions		Package Integrity Leak Test Technologies ⟨1207.2⟩, Package Seal Quality Test Technologies ⟨1207.3⟩

* In case of locally acting API (not systemically absorbed), dissolution rate test may not be done.
** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).
***Particle size distribution testing may be proposed in place of dissolution testing, when development studies demonstrate that particle size is the primary factor influencing dissolution; justification should be provided. The acceptance criteria should include acceptable particle size distribution in terms of the percent of total particles in given size ranges. The mean, upper, and / or lower particle size limits should be well defined
****When the tonicity of a product is declared in its labeling and/or when composition includes tonicity. It is also done for large volume parenteral.

14.Tests performed on tablets:

Test	Applicability		Required Information	Acceptance criteria
	Tablet Type	Done/ Not done		
1.Description: ○ **Appearance** ○ **Colour of tablet**	All		○ Tablet shape ○ Colour (Colour of core & coat in case of coated tablets) ○ Type of coating case of coated tablets ○ Scored or not. ○ Biconvex/flat.	According to manufacturer specifications
2. Mass uniformity* (BP) Procedure according to: Ph. Eur. method 2.9.5	• Type of coat: ○ Uncoated ○ Film coat ○ Sugar coat ○ If the test for uniformity of content is prescribed for all the active substances, the test for uniformity of mass is not required.	○ Yes ○ Yes ○ No		● Not more than 2 of the individual masses deviate from the average mass (actual) by more than the percentage deviation. ● None deviates by more than twice that percentage. (Ph. Eur. method 2.9.5)
3. Disintegration time (Egy.Ph, USP, BP) Procedure according to Disintegration: Egy.Ph (B.3.29) USP-NF ⟨701⟩ Ph. Eur. method 2.9.1 Ph. Eur. method 2.9.2	○ Immediate release ○ Oral lyophilisates ○ Delayed release (enteric coated). ○ Extended release (sustained/modified/controlled).	○ Yes ○ Yes ○ Yes ○ No		Disintegration Egy.Ph (B.3.29) USP-NF ⟨701⟩ (Ph. Eur. monograph 0478) (Ph. Eur. monograph 1164)
4.Dissolution ** Reference of method is chosen from one of the following: ○ Egy.Ph, USP or BP specific monograph*** ○ FDA dissolution methods database. (submit comparative dissolution profile in the most suitable media)**** ○ In-house method Obey the dissolution method development criteria**** Refer to (Annex II)	○ Effervescent tablets that result in a solution ○ Others Disintegration could substitute dissolution as a performance test if a justification submitted by the manufacturer that it obeys the ICH Q6A guidelines. In this case, the performed dissolution method should be supplied by the manufacturer.	○ No ○ Yes	• Dissolution Parameters: ○ Media composition, pH & volume ○ Apparatus type ○ RPM ○ Temperature ○ Sinkers needed (common type: coiled sinker) ○ Filter type (common types: Nylon, PVDF & PTFE) ○ Sampling time(s) ○ Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3

Test	Applicability		Required Information	Acceptance criteria
	Tablet Type	Done/ Not done		
5.Friability (USP & BP) Procedure according to: USP-NF ⟨1216⟩ Tablet Friability BP (Ph. Eur. method 2.9.7)	○ Uncoated ○ Coated	○ Yes ○ No		USP-NF ⟨1216⟩ Tablet Friability BP (Ph. Eur. method 2.9.7)
6. Hardness (USP& BP) Procedure according to Tablet Breaking Force: USP-NF ⟨1217⟩ Ph. Eur. method 2.9.8	○ Uncoated ○ Coated	○ Yes ○ No		According to manufacturer's specifications
7. Subdivision (BP) Procedure according to: Ph. Eur. monograph 0478	○ Functional score. ○ Non-functional score. To skip subdivision test**:** the manufacturer should submit accepted justification. In this case, the word 'Indivisible' should be clearly written on the package. Exceptionally, the package without this word 'Indivisible' could be accepted with a written commitment only in case of pilot batches.	○ Yes ○ No		● NMT 1 individual mass is outside the limits of 85- 115 % of the average mass. ● The tablets fail to comply with the test if more than 1 individual mass is outside these limits, or if 1 individual mass is outside the limits of 75-125% of the average mass. (Ph. Eur. monograph 0478
8.Water content (Egy.Ph, USP) Water determination USP-NF ⟨921⟩ Procedure according to: Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test			According to manufacturer specifications
9.Fineness of dispersion (BP) Procedure according to: Ph. Eur. monograph 0478.	○ Dispersible tablets Others	○ Yes ○ No		A smooth dispersion is produced, which passes through a sieve screen with a nominal mesh aperture of 710 µm.

Test	Applicability		Required Information	Acceptance criteria
	Tablet Type	Done/ Not done		
10. Acid neutralizing capacity (Egy.Ph, USP) Procedure according to Acid neutralizing capacity: Egy.Ph (B.4.1) USP-NF ⟨301⟩	Antacids only			According to manufacturer specifications

* Average weight could be considered if needed as IPC USP-NF (1163) Quality assurance in pharmaceutical compounding.
** In case of locally acting API (not systemically absorbed), dissolution rate test may not be done and disintegration time is sufficient.
*** A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).
**** Reflection paper on the dissolution specification for generic solid oral immediate release products with systemic action EMA/CHMP/CVMP/QWP/336031/2017 may be referred to as guidance.

15. Tests performed on Transdermal Delivery Systems (TDS)

Test	Applicability		Required Information	Acceptance criteria
	TDS type	Done/ Not done		
1. Description	All			According to manufacturer's specifications
2. Dimensions	All			According to manufacturer's specifications
3. Water content (Egy.Ph, USP) Water determination Egy.Ph (B.3.15) USP-NF ⟨921⟩ Procedure according to Manufacturer's method or specific monograph.	If not stated by manufacturer: Need justification to skip test			According to manufacturer's specifications
4. Dissolution Reference of method is one of the following: ○ Egy.Ph, USP or BP specific monograph* ○ FDA dissolution methods database (submit comparative dissolution profile in the most suitable media). ○ In-house method (submit comparative dissolution profile in the most suitable media).	All		• <u>Dissolution Parameters:</u> ○ Media composition, pH & volume ○ Apparatus type ○ RPM ○ Temperature (32˚C) ○ Filter type (common types: Nylon, PVDF & PTFE) ○ Sampling time (at least three, expressed in hours) ○ Q (the amount dissolved)	Dissolution Egy.Ph (B.3.28) USP-NF ⟨711⟩ Ph. Eur. method 2.9.3
5. Particle size	○ Suspension in reservoir ○ Others	○ Yes ○ No		According to manufacturer's specifications
6.Specific Tests for TDS ○ Peel adhesion test ○ Release liner peel test ○ Tack test ○ Cold flow test ○ Shear test	All			According to manufacturer's specifications

* A comparative dissolution profile in the most suitable media is required for old BP monographs (published before 2008).

(Annex II)

1. Development of the in-house dissolution methods

The criteria of dissolution method development and setting dissolution specification of immediate release oral solid dosage forms

Based on the Bio-pharmaceutics classification system (BCS) of the drug substance

BCS class I or III

Containing High Solubility Drug Substances (The dose/solubility volume of solution≤250 ml of aqueous media over the pH range of 1 to 6.8 at 37°C ± 1°C)

Guide I[1]

Or

Guide II[1]

BCS class II or IV

Containing low Solubility Drug Substances

Guide II

Guide I	
Apparatus:	Commonly used: (1) Basket method (USP Apparatus 1) (2) Paddle method (USP Apparatus 2)
Dissolution Medium:	• **The volume of the dissolution medium** is generally 500 mL. (or 900 mL with appropriate justification) • **The composition of the dissolution medium:** o 0.1N HCl in aqueous medium o No surfactant in medium
Temperature	Should be conducted at 37±0.5°C.
Sinkers	In general, capsule dosage forms tend to float during dissolution testing with the paddle method. In such cases, it is acceptable to add a few turns of a wire helix (USP) around the capsule be used.
Agitation	(1) Basket method: 100 rpm. (2) Paddle method: 50 rpm (or 75 rpm with appropriate justification)
Dissolution criteria	Q=80% in 30 minutes.
Applicability	- Solid orally-administered IR drug products, such as tablets and capsules, that are meant to be swallowed. - Does not apply to orally disintegrating tablets (ODT). However, if absorption from the oral cavity can be ruled out for the ODT, then the dissolution of the ODT may fall under this guidance. - Does not apply to sublingual dosage forms which can be ODTs as well. - This guidance may apply to chewable tablets if the dissolution studies are conducted on the intact tablets and the product meets the conditions described in this guidance.

Guide II	
Apparatus:	Commonly used: (1) the basket method (Apparatus 1) (2) the paddle method (Apparatus 2) Described in the USP, and may be considered if needed: (3) reciprocating cylinder (Apparatus 3) and (4) a flow-through cell system (Apparatus 4)
Dissolution Medium:	• **The volume of the dissolution medium** is generally 500, 900, or 1000 mL. Sink conditions are desirable but not mandatory. (2, 4 L or low volume dissolution media in mini vessels need justification). • **The composition of the dissolution medium:** o An aqueous medium with pH range 1.2 to 7.2 (ionic strength of buffers the same as in USP) should be used. To simulate intestinal fluid (SIF), a dissolution medium of pH 6.8 should be employed. A higher pH should be justified on a case-by-case basis and, in general, should not exceed pH 8.0. o To simulate gastric fluid (SGF), a dissolution medium of pH 1.2 should be employed without enzymes. The need for enzymes in SGF and SIF should be evaluated on a case-by-case basis and should be justified. o Gelatin capsule products may need to add enzymes to the dissolution media (pepsin with SGF and pancreatin with SIF) to dissolve pellicles, if formed, to permit the dissolution of the drug. o Use of water as a dissolution medium is discouraged because test conditions such as pH and surface tension can vary depending on the source of water and may change during the dissolution test itself, due to the influence of the active and inactive ingredients. o The need for and the amount of the surfactant should be justified. Use of a hydro-alcoholic medium is discouraged.
Temperature	Should be conducted at 37±0.5°C.
Deaeration	Certain drug products and formulations are sensitive to dissolved air in the dissolution medium will need deaeration.
Sinkers	In general, capsule dosage forms tend to float during dissolution testing with the paddle method. In such cases, it is acceptable to add a few turns of a wire helix (USP) around the capsule be used.
Agitation	o Basket method: 50-100 rpm. (higher than 100 rpm need justification) (Note: Should not exceed 150 rpm) o Paddle method: 50-75 rpm (higher than 75 rpm need justification) (Note: Should not exceed 150 rpm) o Reciprocating cylinder: 5-30 DPM. o Flow through cell: flow rate 4, 8 and 16 mL/min.
Dissolution criteria	According to the comparative dissolution profile in the most suitable media.[1]

[1] Reflection paper on the dissolution specification for generic solid oral immediate release products with systemic action EMA/CHMP/CVMP/QWP/336031/2017.

2. Comparative dissolution study:

Recommendations should be considered in the submitted comparative dissolution studies:

1. The dissolution parameters should be submitted.
2. The dissolution measurements of the test and reference batches should be made under exactly the same conditions.
3. The dissolution time points for both the profiles should be the same (e.g., 15, 30, 45, 60 minutes).
4. A minimum of three time points (zero excluded) should be included.
5. Only one measurement should be considered after 85% dissolution of both the products.
6. In the case where 85% dissolution cannot be reached owing to poor solubility of the API or the release mechanism of the dosage form, the dissolution should be conducted until a plateau has been reached.
7. At least 12 units should be used for determination of each profile.
8. To allow use of mean data, the percent coefficient of variation at the earlier time points (e.g., 10 minutes) should not be more than 20%, and at other time points should not be more than 10%.
9. For curves to be considered similar, f_1 values should be close to 0, and f_2 values should be close to 100. Generally, f_1 values up to 15 (0-15) and f_2 values greater than 50 (50-100) ensure sameness or equivalence of the two curves and, thus, of the performance of the test and reference products.

Annex III

Requirements for Chemical Analysis

1. Active pharmaceutical ingredients (API) used in the manufacture of finished pharmaceutical product (FPP):

1.1 Specifications:

1.1.1 In case the API reference according to the composition is one of the pharmacopoeias; the specifications of the API in the certificate of analysis should follow that pharmacopoeia.

1.1.2 In case of in-house API:

1.1.2.1 If the API has a monograph in any of the pharmacopoeias, specifications of the API supplier are accepted only if they comply with the specifications listed in the pharmacopoeia or tighter specifications.

1.1.2.2 If the API doesn't have any monographs in any of the pharmacopoeias, specifications of supplier are accepted provided the following:

- Tests for impurities will be evaluated according to ICH Q3A guidelines for impurities.
- For API present as both a chiral single enantiomer and as racemate, identity testing(s) for verification of chirality is more appropriately addressed as part of the drug substance specification.

2. Finished pharmaceutical products (FPP): CADC laboratories

2.1 Specifications and Certificate of Analysis:

2.1.1 For products described as pharmacopeial, specifications of this product must follow the specifications in the whole monograph in the reference pharmacopoeia.

2.1.2 For products that have pharmacopeial monograph(s), specifications listed in the pharmacopeial monograph are used as the main reference in the evaluation of the required tests and specifications.

2.1.3 Identification tests:

2.1.3.1 Identification test item must be included in the specification sheet and finished product certificate of analysis (CoA)

2.1.3.2 Titrimetry is not an identification test

2.1.3.3 Identification test should be applied by at least two techniques.

2.1.4 Assay of API, antimicrobial preservatives and antioxidants:

2.1.4.1 Limits for assay should be expressed in terms of active moiety (free acid or base, anhydrous basis) unless otherwise specified in the specific monograph

2.1.4.2 The general acceptance limits are as follows:

- General acceptance limit for the API is 90-110% of the Labeled claim.
- General acceptance limit for the preservative is 80-120% of the Labeled claim
- General acceptance limit for the antioxidant is according to manufacturer specification with scientific justification.

2.1.4.3 In case of approved stability overage where the limit of assay in such a case will be 90% of labeled claim to 110% of labeled claim +overage (approved in composition as stability overage).

2.1.4.4 Analysis of preservatives in solid dosage form in capsule shells is not mandatory unless it is listed in the manufacturer specifications.

2.1.4.5 Analysis of any other excipients is not mandatory unless it is listed in the manufacturer specifications.

In all cases deviation (wider) from general acceptance limit may be accepted only if justified by:
1- Specific monograph for the FPP.
2- Approved stability specifications.
Narrower limits are always accepted as manufacturer specifications.

2.1.5 Test of impurities

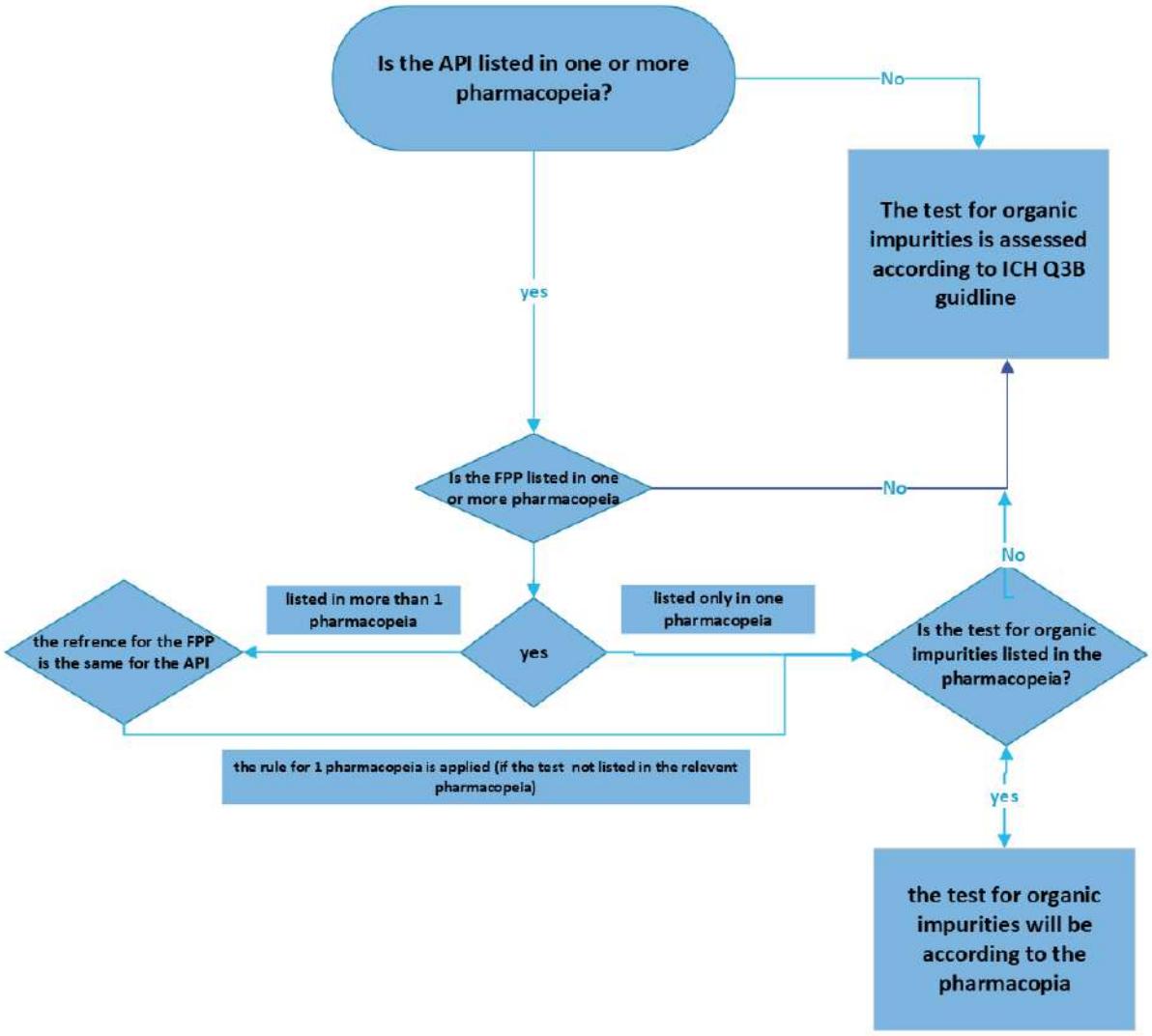
2.1.5.1 Organic impurities/ related substances:

- In USP monographs of capsules, the definition does not specify the type of capsule (gelatin, Hypromellose, starch derivative, hard, soft, etc.), or the type of filling in the capsule (powder, granules, pellets, liquid, semisolid, etc.). Accordingly, the test for organic impurities described under the monograph, if present, must be applied to any of the previously mentioned types.
- In USP monographs of tablets, unless otherwise stated, the tablets are considered immediate release regardless of the coat and shape of the tablets (film coated, sugar

coated, caplets.) and test of organic impurities described under the USP monograph, if present must be applied.

- Same decision tree will be followed in case of presence of more than one API.

Decision tree for organic impurities:



2.1.5.2 Residual solvents:

- Assessment of residual solvents impurities will be according to ICH Q3C, unless otherwise specified in the specific monograph.
- Analytical procedures for the determination of solvent classes can be followed as described under Egy.Ph (B.2.12) or USP < 467>.
- Alternative validated methodologies may also be used or modifications to the official methods may be done to demonstrate compliance with the defined limits where verification of Egy.Ph and USP procedures or validation of alternative methods for residual solvents is performed according to Egy.Ph (B.2.12) or USP<1467>.
- If the residual solvent is controlled in an intermediate (like granules or pellets) by analysis in CADC raw material laboratories, reanalysis will not be necessary in the finished product but should be included in finished product specifications.

2.1.5.3 Elemental impurities

- Risk assessment for elemental impurities according to ICH Q3D should be conducted for Group II products, submitted for new MA approval.
- If the total elemental impurity level from all sources in the drug product is expected to be consistently less than 30% of the PDE, then additional controls are not required
- If the limit is defined as 30% of the PDE of the specific elemental impurity under consideration, a control strategy may be required to ensure that the PDE is not exceeded in the drug product.

2.1.5.4 Mutagenic impurities

- Mutagenic impurities should be controlled according to ICH M7 for Group II products, submitted for new MA approval
- EDA guideline for control of nitrosamine impurities should be followed

2.1.6 Uniformity of dosage unit:

To ensure the consistency of dosage units, each unit in a batch should have drug substance content within a narrow range around the label claim. Dosage units are defined as dosage forms containing a single dose or a part of a dose of drug substance in each unit. The uniformity of dosage unit's specification is not intended to apply to solutions, suspensions,

emulsions, or gels in unit-dose containers intended for local action following external, cutaneous administration.

2.1.6.1 CADC laboratories will use the interchangeable general chapter of the Uniformity of dosage units Egy.Ph (B.3.32), USP <905>, Ph.Eur. 2.9.40. and JP 6.02 where target Value (T) =100% otherwise stated in the product monograph.

2.1.6.2 Where different procedures are used for assay of the preparation and for the Content Uniformity test, it may be necessary to establish a correction factor to be applied to the results of the latter.

2.1.6.3 CADC laboratories will apply; whenever applicable; the method of assay for the determination of API(s) in the evaluation of content uniformity test in case the method of content uniformity is not submitted.

2.1.7 Alcohol content.

For liquid formulation contains a quantity of alcohol this test will be evaluated according to Egy.Ph (B.3.39) or USP <611>.

2.2 Method of analysis (MOA):

A specific, stability-indicating assay method to determine strength (content) should be included for all drug products.

In cases where use of a non-specific assay is justified, other supporting analytical procedures should be used to achieve overall specificity. For example, where titration is adopted to assay the drug substance for release, the combination of the assay and a suitable test for impurities can be used.

2.3 Method Validation and Verification:

2.3.1 When a non-pharmacopeial method is used a full validation study must be submitted with the method of analysis. Validation will be assessed according to ICH Q2 (R2).

2.3.2 When official pharmacopeial analytical methods are applied out of their intended scope according to the description stated in the pharmacopeial monograph (e.g. method for API(s) to be applied on finished products, finished product of different dosage forms, or in presence of other API (s), full validation study will be essentially required to be submitted for the applied analytical method.

2.3.3 When a pharmacopeial method is used, verification is performed according to Egy.Ph (D.56) or USP <1226> and/or OMCL guidelines.

2.4 Analysis requirements:

2.4.1 Standards:

2.4.1.1 A certified reference material (CRM), pharmacopeial or otherwise (any accredited reference standard), is preferable.

2.4.1.2 In case a working standard is submitted, EDA template for COA of a working standard is mandatory, and the lot number for the primary standard used in its qualification, as evidence of traceability, must be stated in the submitted COA.

2.4.2 Analytical Columns:

The use of equivalent columns is accepted if within permissible limits according to Egy.Ph (B.3.1) or USP <621> and the verification on the new chromatographic conditions is required.

2.4.3 Placebo:

Placebo should be provided in case of organic impurities testing. If the placebo is unavailable the company should send a declaration of acceptance to start the analysis of impurities without placebo and will be committed to provide it with other analysis requirements and reference standards in case the analysis gives unsatisfactory results.

2.5 Special considerations:

2.5.1 Benzalkonium chloride:

The presence of at least Benzalkonium chloride homologs c12 and c14 is mandatory for confirmation of identification of Benzalkonium chloride and the submitted method of analysis must be able to discriminate Benzalkonium chloride homologs.

2.5.2 For products used as sources of elements &/or minerals:

2.5.2.1 Identification:

The identification testing is needed for either the salt itself or the individual ions composing it according to the latest pharmacopoeia and in case of complexes such as iron dextran, iron polymaltose, iron sucrose ……etc.,

detailed identification method for both the cation (e.g. iron) & organic moiety should be attached.

2.5.2.2 Assay:

It is accepted for the salt itself or the cations (Na^{+}, K^{+}, Ca^{++}, Mg^{++}, Cu^{++}, Mn^{++}, Se^{3+}, Cr^{3+}, Mo^{+}, Zn^{++}, Fe^{++}, B^{++}, $Bi^{3+,}$ P^{4+}) and/or the anions (Citrate, acetate, chloride, oxalate, lactate, carbonate, bicarbonate, fluoride and iodide)

2.5.2.3 For limits of assay, pharmacopeial acceptance criteria are generally applied whenever available.

Annex IV

Requirements for Microbiological Analysis

This section outlines the requirements and test specifications for the following microbiological tests:

1. Microbiological examination of non-sterile products.
2. Probiotics tests.
3. Sterility test (For sterile dosage forms).
4. Antibiotic potency testing (Microbiological assay method).
5. Bacterial endotoxin test.
6. Disinfectant efficacy testing.

General instructions:

- CADC reserves the right to request analysis resources (e.g., reference strains, reference standards, culture media, canisters, neutralizers, etc.) as needed.
- The analysis will be conducted in accordance with the most recent pharmacopoeia editions.

1. Microbiological examination of non-sterile products (Microbial enumeration test & Test for specified microorganisms)

The tests are primarily intended for evaluating whether a substance or preparation complies with predetermined standards for microbiological quality.

1.1 <u>The following information is required:</u>

1.1.1 Test specifications:

The following should be provided.

Tested parameters	e.g., Total aerobic microbial count (TAMC), Total combined yeasts and molds count (TYMC), Tests for specified microorganisms

Nature of the product	Water-soluble products, non-fatty products insoluble in water, fatty products, fluids or solids in aerosol form, or transdermal patches
Route of administration	Non-aqueous preparations for oral use, aqueous preparations for oral use, rectal use, Oromucosal use, gingival use, nasal use, cutaneous use, auricular use, transdermal patches, vaginal use, inhalation use
Used technique	*For microbial enumeration tests:* Membrane filtration, Plate-count methods, or Most-Probable-number method) *For test method for specified microorganisms:* test for *Escherichia coli*, test for *Salmonella sp.*, test for bile tolerant Gram-negative bacteria, test for *Pseudomonas aeruginosa*, test for *Staphylococcus aureus*, test for Clostridia, test for *Candida albicans*, and/or test for *Burkholderia cepacia* complex
Category of the used botanical ingredients*	chopped or powdered botanicals, botanical extracts, tinctures, infusions, decoctions, fluid extracts, botanicals to be treated with boiling water before use
Results and acceptance criteria	Complies with specification Should be expressed in cfu/g, cfu/ml, cfu/patch, or cfu/film
References	Egy.ph, USP, BP, European pharmacopoeia

1.1.2 Acceptance criteria:

Unless more stringent criteria are specified in the FPP monograph, the recommended acceptance criteria for various pharmaceutical preparations are given in **Table 1** and **Table 2 (a and b)**.

***For a given preparation that contains more than one API and each API has a specific individual monograph, the preparation will be tested using less stringent criteria for microbial enumeration tests, whilst the more stringent criteria will be applied on tests for specified microorganisms to ensure the microbiological quality of the product.**

When the acceptance criterion for microbiological quality is prescribed, it is interpreted as follows:

10^1 cfu: maximum acceptable count = 20,

10^2 cfu: maximum acceptable count = 200,

10^3 cfu: maximum acceptable count = 2000; and so forth.

Table 1: Acceptance criteria for non-sterile dosage forms and non-sterile substances for pharmaceutical use (Harmonized between USP and EP):

Route of administration	TAMC (cfu/g or cfu/ml)	TYMC (cfu/g or cfu/ml)	Specified microorganism(s)
Nonaqueous preparations for oral use	10^3	10^2	Absence of *Escherichia coli* (1g or 1ml)
Aqueous preparation for oral use	10^2	10^1	Absence of *Escherichia coli* (1g or 1ml) *Burkholderia cepacia* complex (1g or 1ml)
-Oro mucosal use, **-Gingival use,** **-Nasal use,** **-Cutaneous use,** **-Auricular use** **-Transdermal patches (limits for one patch including adhesive layer and backing)**	10^2	10^1	Absence of *Staphylococcus aureus* (1g, 1ml, or patch) *Pseudomonas aeruginosa* (1g, 1ml, or patch)
Aqueous preparation for -Oro mucosal use, nasal use, or cutaneous use.	10^2	10^1	Absence of *Staphylococcus aureus* (1g, 1ml, or patch) *Pseudomonas aeruginosa* (1g, 1ml, or patch) *Burkholderia cepacia* complex (1g or 1ml)
Oro dispersal films acc. to EP	10^2	10^1	Absence of *Escherichia coli* (1 film) *Staphylococcus aureus* (1 film) *Pseudomonas aeruginosa* (1 film)
Inhalation use acc. to USP (Non-sterile) (Special requirements apply to liquid preparations for	10^2	10^1	Absence of *Staphylococcus aureus* (1g or 1ml) *Pseudomonas aeruginosa* (1g or 1ml)

nebulization)			Bile tolerant Gram-negative bacteria (1g or 1ml) *Burkholderia cepacia* complex (1g or 1ml)
Inhalation use acc. to EP (Non-sterile) (Special requirements apply to liquid preparations for nebulization)	10^2	10^1	Absence of *Staphylococcus aureus* (1g or 1ml) *Pseudomonas aeruginosa* (1g or 1ml) Bile tolerant Gram-negative bacteria (1g or 1ml)
Vaginal use	10^2	10^1	Absence of *Staphylococcus aureus* (1g or 1ml) *Pseudomonas aeruginosa* (1g or 1ml) *Candida albicans* (1g or 1ml)
Rectal use	10^3	10^2	---
Oral dosage forms containing raw materials of natural origin (animal, vegetal or mineral) (TAMC of raw material > 10^3 cfu/g or ml) acc. to EP.	10^4	10^2	Absence of *Staphylococcus aureus* (1g or 1ml) *Escherichia coli* (1g or 1ml) *Salmonella spp.* (10g or 10ml) Bile tolerant Gram-negative bacteria (NMT 10^2 cfu /1g or 1ml)
Substances for pharmaceutical use	10^3	10^2	The assessment takes account of the processing to which the substance is subjected

Table 2: Acceptance criteria for non-sterile nutritional and dietary supplements and dietary supplements ingredients

a) **According to USP**

Herbal medicinal products containing	TAMC (cfu/g or cfu/ml)	TYMC (cfu/g or cfu/ml)	Specified microorganism(s)
Dried or powdered botanicals	10^5	10^3	Absence of *Salmonella spp.* and *E. coli* in 10g Bile tolerant Gram-negative bacteria (NMT 10^3 cfu/g or ml)
Powdered botanical extracts, Nutritional supplements with botanicals	10^4	10^3	Absence of *Salmonella spp.* and *E. coli* in 10 g
Tinctures, Fluid extracts	10^4	10^3	---
Infusions/decoctions	10^2	10	---
Botanicals to be treated with boiling water before use	10^6	10^4	Absence of *E. coli* in 10 g *Salmonella* in 10 g Bile tolerant Gram-negative bacteria (NMT 10^2 cfu /g or ml)
Raw materials and Dietary supplement ingredients	10^3	10^2	Absence of *E. coli* in 10 g
Nutritional supplements with synthetic or highly refined ingredients	10^3	10^2	Absence of *E. coli* in 10 g

b) **According to European Pharmacopoeia:**

Herbal medicinal products containing	TAMC (cfu/g or cfu/ml)	TYMC (cfu/g or cfu/ml)	Specified microorganism(s)
Herbal drugs, intended for the preparation of infusions and decoctions using boiling water (for example herbal teas)	10^7 Max. acceptable count: 50 000 000	10^5 Max. acceptable count: 500 000	*E. coli* 10^3 cfu/g Absence of *Salmonella* in (25 g)

Herbal medicinal products containing	TAMC (cfu/g or cfu/ml)	TYMC (cfu/g or cfu/ml)	Specified microorganism(s)
Extracts and/or herbal drugs, where the method of processing or of pre-treatment reduces the levels of organisms to below those stated for this category	10^4 Max. acceptable count: 50 000	10^2 Max. acceptable count: 500	Absence of *E. coli* in 1 g or 1 ml *Salmonella* in 25 g or 25 ml Bile tolerant Gram-negative bacteria (NMT 10^2 cfu/g or cfu/ml)
Extracts and/or herbal drugs, where the method of processing or pre-treatment, would not reduce the level of organisms sufficiently (for example, extraction with low-strength ethanol or water that is not boiling, or low-temperature concentration)	10^5 Max. acceptable count: 500 000	10^4 Max. acceptable count: 50 000	Absence of *E. coli* in 1 g or 1 ml *Salmonella* in 25 g or 25 ml Bile tolerant Gram-negative bacteria (NMT 10^4 cfu /g or cfu/ml)

1.1.3 Method suitability certificate: especially for products with proved antimicrobial activity or if insufficient information about the product exists to judge its probable growth inhibiting activity.

This certificate proves the suitability of the chosen method for microbial enumeration test and detection of specified microorganisms in the presence of the product to be tested, particularly for products with proven antimicrobial activity or if insufficient information about the product exists to judge its probable growth inhibiting activity.

1.1.4 Reduced frequency of microbial testing

Pharmaceutical drug products with water activities well below **0.75** (e.g., direct compression tablets, powder and liquid-filled capsules, non-aqueous liquid products, ointments, and rectal suppositories) would be excellent candidates for reduced microbial limit testing.

In order to obtain approval for reduced frequency of microbial testing or skipped lot testing or eliminate routine testing; the applicant should introduce the following (USP 44-chapter 1112):

○ Formulation of the drug product has antimicrobial properties (as antibiotics) acc. to Antimicrobial effectiveness testing USP chapter <51> or it does not support microbial growth or viability (i.e: with low water activity).
○ Proof that the product has been manufactured from ingredients of good microbial quality.
○ Demonstrated effectiveness of microbial contamination control of the raw material, ingredient water, manufacturing process, formulation, and packaging system that prevent moisture.
○ Proof that manufacturing sites have an established testing history of low bioburden associated with their products.
○ Historic testing database of the product; the testing history would include microbial monitoring during product development and routine testing of sufficient marketed product lots (e.g. up to 20 lots) to ensure that the product has little or no potential for microbial contamination.

2. Probiotic Tests

Probiotics are live microorganisms that confer health benefits to the host when administered in adequate amounts. This section provides required information for identification, enumeration, contamination, and other probiotic requirements.

2.1 The following information is required

2.1.1 Test specifications

The following information should be provided:

The dosage form (intended users)	Tablets, capsules, soft gels, gel caps, powders, chewable gels (adult or infant use)
Probiotic classification	Non-spore-forming bacteria, spore-forming bacteria, or yeast and molds
Microbial strain(s) used	Lactobacillus, Bifidobacterium, Saccharomyces, etc.
Identification method	Polymerase chain reaction (PCR) with strain-specific primers is recommended.
Enumeration method and count of strains	Results should be expressed as viable colony forming units (cfu) per gram
Test for contaminant microorganism(s) **Test for non-lactic acid bacteria** **Test for total yeast and molds**	Result should be expressed as viable colony forming units (cfu) per gram
Absence of pathogenic microorganisms	As specified in the individual monograph
Method suitability	Must be established and provided
References	USP

2.1.2 Acceptance criteria for probiotics

Unless otherwise specified in the individual monograph, the recommended acceptance criteria for probiotics or finished product for oral use are as follows:

Probiotic class	Non-lactic acid-bacteria (cfu/g)	TYMC (cfu/g)	Specified microorganism(s)
Non-spore-forming bacteria	NMT 5×10^3	NMT 1×10^2	Absence of E. coli and Salmonella spp. in 10 g
Spore-forming bacteria	--	NMT 1×10^2	Absence of E. coli and Salmonella spp. in 10 g
yeast and molds	--	NMT 1×10^3	Absence of E. coli and Salmonella spp. in 10 g

- The absence of *Clostridium perfringens* and *Cronobacter sakazakii* in addition to the absence of the listed specified microorganisms should be tested and confirmed if a probiotic ingredient or finished product is intended for infant use.
- The absence of *Listeria monocytogenes*, *Staphylococcus aureus*, or *Pseudomonas aeruginosa* in addition to the absence of the listed specified microorganisms should be tested and confirmed if a probiotic ingredient or a finished product containing a probiotic ingredient(s) poses a risk associated with the contamination of the above microorganisms

N.B. Preparations containing killed organisms or their metabolites are not considered probiotics and can be categorized as dietary and nutritional supplements (refer to Microbiological examination of non-sterile products (Microbial enumeration test & Test for specified microorganisms) section.

3. Sterility test

The test is applied to substances, preparations, or articles that, according to Pharmacopoeia, are required to be sterile. However, a satisfactory result only signifies the absence of any contaminating microorganism in the sample under test conditions.

3.1 Test specifications

The following specifications should be provided:

Tested parameter	Sterility of the product
Used technique	Membrane filtration or direct inoculation of the culture media
Nature of the product	Aqueous solutions, soluble solids, solids, oils, oily liquids, ointments, or creams
Acceptance criteria	Compliance with the test for sterility
References	Egy.ph, USP, BP, European Pharmacopoeia

3.2 Certificate of method suitability

This certificate proves the ability of the chosen sterility testing method to detect microorganisms in the presence of the product to be tested.

4. Antibiotic Potency Testing

The antibiotic assay is applied to determine the potency of antibiotics by evaluating their inhibitory effects on specific microorganisms under controlled conditions. Microbiological assays are sensitive to changes in antimicrobial activity that might not be detectable through chemical methods alone. The microbiological assay of antibiotics should be carried out using standardized procedures such as the cylinder-plate (agar diffusion) method and/or the turbidimetric method.

General notes:

1. For Pharmacopoeia API and finished products whose monographs specify the assay of antibiotics is by HPLC or a chemical assay, it is NOT ACCEPTED to submit an in-house microbiological assay.
2. For finished products comprising antibiotic combinations that do not have an individual monograph, the in-house method should include an approach for separating the component antibiotics to determine the individual potency for each antibiotic component.
3. Specific considerations for non-Pharmacopoeia finished products:
    a. In cases where the dosage form of a product does not have a separate monograph, but the Pharmacopoeia provides a microbiological assay method for the API, the analytical procedure and acceptance criteria will be applied following the general chapter.
    b. In cases where neither the dosage form of the product nor the API is covered by a Pharmacopeial microbiological assay method, an in-house method should be provided. This method should be validated as per the performance characteristics stated in Pharmacopoeia, such as Egy.Ph (D.55) or USP <1225>, including method linearity, accuracy, precision, ruggedness, and robustness.

a. Test specifications

Tested parameter	Potency of antibiotics
Technique	Cylinder Plate Method (Agar diffusion) Turbidimetric Method
Reference Standard Potency expressed in IU/mg or µg/mg	• Reference Standard, Certified Reference Material, certificate and validity of reference standard should be submitted with storage conditions, Safety Data Sheet (SDS) • In case of using Working Standards, the applicant should submit specifications, complete data of standardization, reference of the technique and the acceptance criteria, traceability, and reference standards used in standardization
Test organism	As indicated in the used reference. Please mention the ATCC number of the reference strain used, inoculum preparation, and standardization
Procedure	• Procedure for preparations of initial, final, and median concentrations for both reference standard and tested antibiotic • Initial solvents, further and final diluents • Buffers used with their preparation procedure • Incubation conditions, culture media, temperature, and incubation time
Calculations	• Detailed equation should be submitted with the definition of each parameter according to the reference used (USP, BP, Ph. Eur.) • Excel sheet copy (on demand)
Acceptance criteria	According to the reference used
Reference	Egy.ph, Ph. Eur., BP, or USP • Copies of the non-compendial analytical procedures used to generate testing results should be provided. • Unless modified, it is not necessary to provide copies of the compendial analytical procedures.

c. Bacterial Endotoxin Test (BET)

The BET is an *in vitro* assay for detection and quantitation of bacterial endotoxins, a component of the cell wall of Gram-negative bacteria. The test is performed on injectable pharmaceutical products that come into direct or indirect contact with the circulatory, lymphatic, or cerebral fluid systems. **The following information are required:**

d. MVD* calculation

$$Max.\ Valid\ Dilution\ (M.V.D) = \frac{Endotoxin\ limit\ X\ product\ conc.}{Lysate\ sensitivity\ (\lambda)}$$

e. Test specifications

Tested parameter	Bacterial endotoxin limit (BEL)
Detailed method of analysis	Inhibition/Enhancement test is highly recommended with any special precautions
Reference used	(USP, BP, or Ph. Eur.)
Calculation of BEL (K/M)	In case of non-Pharmacopeial products
Pamphlet of the product	If unavailable, then the provision of the pamphlet of the reference product is recommended

f. Acceptance criteria:

Route of administration	Bacterial Endotoxin Limit (BEL)		
	Pharmacopeial products According to (USP, BP, or Ph. Eur.)	**Non-Pharmacopeial products** (Calculate *BEL= K/M*)	
		K (the max. pyrogenic dose/Kg), (Constant depends on RoA)	**M** (the max. recommended dose /Kg)
Intravenous (IV) for parenteral products	As stated in the specific monograph of each product	5 EU/kg of body weight	Maximum dose per kilogram administered in 1 h
IV for radiopharmaceuticals		175 EU	Volume of the maximum recommended dose
Intrathecal (IT) for parenteral products		0.2 EU/kg of body weight	Maximum dose per kilogram administered in 1 h
IT for radiopharmaceuticals		14 EU	Volume of the maximum recommended dose
Parenteral administered per square meter of body surface ***(USP)***		100 EU/m^2	Maximum dose per square meter per hour
Injections other than IV (intramuscular, subcutaneous, etc.)		5 EU/kg of body weight	Maximum dose per kilogram administered in 1 h
Intraocular fluids ***(USP)***	As stated in the specific monograph of each product	0.2 EU/mL	----------
Anterior segment solid devices ***(USP)***		0.2 EU/device	----------
Ophthalmic irrigation products ***(USP)***		0.5 EU/mL	----------
Injected or implanted ophthalmic drug product ***(USP)***		2 EU/dose	----------

Notes:
- The Chosen dose should be the greatest recommended dose for the lowest body weight in the targeted patient population (**take into consideration the recommended doses for pediatrics**).

Rabbit Test

Rabbit test is only accepted in case of products incompatible with LAL techniques due to interference. Complete justification that proves the incompatibility must be submitted with its supportive results

a-Test Specifications

1)Tested parameter	Testing for pyrogens
2)Detailed method of analysis	Detailed SOP of each product must be submitted containing at least the following: o Diluent used in case of powder products/materials which will be reconstituted. o Dose to be administrated per Kg. o Dose preparation.
3) Reference	e.g.*: Ph. Eur.-General chapter (2.6.8)*

b- Acceptance criteria

No rabbit shows an individual rise in temperature of 0.5 C° or more above its respective control temperature to meet the requirements for the absence of pyrogen, or:

Any other mentioned criteria according to the used reference.

6. Disinfectants Efficacy Testing

The test confirms that disinfectants and antiseptics are active against test organisms, including bacterial and fungal strains, under experimental conditions that simulate their practical use.

a. Test specifications

Tested parameter	Disinfectant efficacy test
Test method	Quantitative suspension method
Neutralizing agents	Will be chosen based on the chemical composition of the disinfectant.

b. Additional information required

Chemical composition of disinfectant	i.e. aldehydes, alcohols, phenolic, quaternary ammonium compounds, etc.
Classification or intended use	General purpose disinfectant, bactericidal, fungicidal, or sporicidal agent.
Directions for use	This should be addressed in the labeling, including the suggested concentrations.

c. Strains used for testing disinfectant activity

Strains for bactericidal activity testing	-Staphylococcus aureus such as ATCC 6538 -Enterococcus hirae such as ATCC 10541 -Escherichia coli such as ATCC 8739, ATCC 11229 - Pseudomonas aeruginosa such as ATCC 15442, ATCC 9027
Strain for yeasticidal activity testing	-Candida albicans such as ATCC 10231
Strains for fungicidal activity testing	-Candida albicans such as ATCC 10231 -Aspergillus brasiliensis such as ATCC 16404
Strains for sporicidal activity testing	Bacillus subtilis such as ATCC 19659, ATCC 6633 Bacillus cereus ATCC 12826

d. Interpretation and acceptance criteria

According to the product composition and the mode of action of active ingredient

Bactericidal activity capability of a product to reduce the number of viable bacterial cells of relevant test organisms	≥ 5 log reduction
Fungicidal activity capability of a product to reduce the number of viable yeast cells and mold spores of relevant test organisms	≥ 4 log reduction
Yeasticidal activity capability of a product to reduce the number of viable yeast cells of relevant test organisms	≥ 4 log reduction
Sporicidal activity capability of a product to reduce the number of viable bacterial endospores of relevant test organisms	≥ 4 log reduction

Annex V

Method Validation and Verification

1. As per ICH Q2 (R2) guideline, the submitted validation study should provide sufficient evidence to demonstrate that the analytical procedure is suitable for its intended purpose, through validation of the relevant performance characteristics of the procedure, using appropriate validation tests, to ensure the quality of the measured result.

Measured Quality Attribute / Analytical Procedure Performance Characteristics to be Demonstrated (2)	IDENTITY	IMPURITY (PURITY) Other quantitative measurements (1)		ASSAY Content or potency / Other quantitative measurements (1)
		Quantitative Test	Limit Test	
Specificity (3) Specificity Test	+	+	+	+
Range				
Response (*Calibration Model*)	-	+	-	+
Lower Range Limit	-	QL†	DL	-
Accuracy (4)				
Accuracy Test	-	+	-	+
Precision (4)				
Repeatability Test	-	+	-	+
Intermediate Precision Test	-	+ (5)	-	+ (5)

- signifies that this test is not normally conducted

+ signifies that this test is normally conducted

† in some complex cases DL may also be evaluated

QL, DL: quantitation limit, detection limit

(1) other quantitative measurements can follow the scheme for impurity, if the range limit is close to the DL/QL; other quantitative measurements can follow the scheme for assay (content or potency), if the range limit is not close to the DL/QL

(2) some performance characteristics can be substituted with technology-inherent justification in the case of certain analytical procedures for physicochemical properties

(3) lack of specificity of one analytical procedure should be compensated by one or more other supporting analytical procedures, unless appropriately justified

(4) alternatively, a combined approach can be used to evaluate accuracy and precision

(5) where reproducibility has been performed and intermediate precision can be derived from the reproducibility data set, an independent study for intermediate precision is not required

Figure 1. Typical performance characteristics and related validation tests for measured quality attributes -ICH Q2(R2)

Table 1 Acceptance criteria for validation parameters of analytical methods employed in quantitative analysis of drug product quality characteristics

Validation Items	Requirement	Acceptance Criteria
Specificity	Blank measurement	Resolution: NLT 2 / Blank interference NMT 1%
	Placebo measurement	Resolution: NLT 2 / Placebo interference NMT 2%
	Peak purity	of lack of interference according to Demonstration software used
	Spiking with potential impurities	Resolution between the target analyte and adjacent peaks NLT 2
	Degradation under stress condition	No indication of another peak under the API peak (Resolution ≥ 2) in degraded solution of API under various stress conditions (Hydrolytic, oxidative, thermal, photolysis).
Range **Response**	Minimum five standard solutions covering: o 80-120% **(assay)** o 70-130% **(content uniformity)** o Reporting level - 120% of specifications **(impurities)**	 R2 ≥ 0.995 (For Assay) R2 ≥ 0.99 (For impurities)
Lower range limit **LOD** **(limit of detection)**	The lowest amount of analyte in a sample which can be detected but not necessarily quantitated	Signal to noise ratio (S/N) ≥ 3:1.
LOQ **(limit of quantification)**	The lowest amount of analyte in a sample which can be quantitatively determined with suitable precision and accuracy.	Signal to noise ratio (S/N) ≥ 10:1.
Precision **Repeatability**	Minimum of nine determinations covering the specified range for the procedure (i.e., three concentrations and three replicates of each concentration) or using a minimum	For Assay RSD ≤ 3% For Impurities: Level < 0.1%, RSD ≤ 30%, n ≥ 6 Level 0.1% - 0.2%, RSD≤ 20%, n ≥ 6

Validation Items	Requirement	Acceptance Criteria
	of six determinations at 100% of the test concentration	Level 0.2 - 0.5%, RSD ≤10%, n ≥ 6 Level 0.5 - 5%, RSD ≤ 5%, n ≥ 6
Intermediate Precision	Expresses within laboratories variations: different days, different analysts, and different equipment.	RSD ≤ 3% Assay
Accuracy	Matrix spiked at 3 levels covering linearity range (nine determinations) (i.e., three concentrations and three replicates of each concentration) (n=9)	95-105.0% Assay for impurities: Level ≤ 0.2%: 70–130% 0.2–0.5%: 80–120% Level 0.5–5%: 90–110%
System suitability	100% concentration of standard solution	Otherwise specified in specific monograph: System repeatability n=5; RSD NMT 2% System Sensitivity S/N ratio ≥ 10:1 for impurities) Resolution R ≥ 2 Tailing factor 0.8-1.8 Theoretical plates ≥ 2000 Capacity factor K ≥ 2

Robustness testing should show the reliability of an analytical procedure in response to deliberate variations in *analytical procedure parameters*, as well as the stability of the sample preparations and reagents for the duration of the procedure, if appropriate. The robustness evaluation can be submitted as part of development data for an analytical procedure on a case-by-case basis or should be made available upon request.[2]

[2] ICH Q2(R2)

Table 2 Acceptance criteria for validation parameters of analytical methods employed in quantitative analysis of dissolution

Validation Items	Requirement		Acceptance Criteria
1- Specificity	Demonstrate the absence of interferences of the following: o Placebo.		o Should not exceed 2%
	o Dissolution media.		o Should not exceed 1%
	o Other active drug substances & Degradants		o Should not exceed 2%
2- Range **Response**	A minimum of 5 concentrations is recommended		o The Y-intercept should not be significantly different from zero. o R^2 should be ≥ 0.98.
	Immediate release One point specification Q – 45% of the lowest strength Multiple point specification Lower limit of reportable range (as justified by the specification) or QL, as appropriate Modified release Lower limit of reportable range (as justified by the specification) or QL, as appropriate	to 130% of declared content of the highest strength	

Validation Items	Requirement	Acceptance Criteria
4- Precision **Repeatability**	○ A minimum of 9 determinations covering the specified range for the procedure (e.g. 3 concentrations/ 3 replicates each) or ○ A minimum of 6 determinations at 100% of the test concentration.	○ %RSD < 2%
Intermediate Precision[3]	○ Typical variations to be studied include days, analysts, equipment, etc. "At least 2 different analysts on 2 days"	○ The difference in the mean value for dissolution results between any two conditions does not exceed an absolute 10% at time points with <85% dissolved and does not exceed 5% for time points NLT 85%. ○ Acceptance criteria may be product specific, and other statistical tests and limits may be used
7- Accuracy & Recovery	○ Accuracy should be assessed using a minimum of 9 determinations over a minimum of 3 concentration levels covering the specified range (e.g. 3 concentrations/ 3 replicates each of the total analytical procedure). ○ Accuracy should be reported as percent recovery.	○ Recovery percentage should be between 95% - 105%

Reproducibility is assessed by means of an inter-laboratory trial. Investigation of reproducibility is usually not required for regulatory submission but should be considered in cases of standardization of an analytical procedure, for instance, for inclusion of procedures in pharmacopoeias and in cases where analytical procedures are conducted at multiple sites.

[3] In cases where reproducibility has been performed, intermediate precision is not needed

Annex VI

Required Documents for File Submission to CADC

Group 1	Products which are either locally produced or imported, that have previously received a registration license, submitted to the Administration of Evaluation and Approval for laboratory testing, or submitted to the Administration of Post Approval Control laboratory testing for purposes other than post approval changes/variations.
Group 2	Products which are either locally produced or imported, submitted to o The Administration of Evaluation and Approval, for MA, re-registration, post approval variations or analytical file renewal o The Administration of Post Approval Control, for post approval variations.

Group 1 Files

The file consists of the following:

- **Regulatory folder**
- **Technical quality folder**

Regulatory folder contains the following

1. Sample analysis request and sample collection report, and renewed report if present.
2. Registration license and other relevant approvals (eg. variation approvals)
3. Copy of the Final Report of analysis, issued by CADC for registration of the product, and in case it is not available, Group I general rules, subclause 6.4.1, shall be followed.
4. Copy of the stability studies approval, if available.
5. Formulation composition according to which product has been manufactured

Technical quality folder file consists of the following:

1. Certificate of analysis of batches, sub batches.
2. Method of analysis and validation in case of changing from registration file and receipt for this change
3. Certificate of analysis and supplier's specifications for reference standard

Group 2 Files

The file consists of the following:

- **Regulatory Folder**
- **Technical Folder**

Regulatory folder contains the following:

1. Registration license and other relevant approvals (eg. variation approvals)
2. Approval letter from Central Administration of Pharmaceutical Products (CAPP), for production batches submitted to CADC before receiving a registration license.
3. Registration Form
4. Quantitative composition
5. Sample analysis request and sample collection report, and renewed report if present.
6. COA of APIs
7. Finished product COA.
8. Box approval
9. Payment receipt
10. Material safety data sheet for all API and anti-oxidant and preservative
11. A declaration, by the applicant, of the category of sample collection and the number of batches sampled
12. Declaration that the information in the file submitted for assessment is correct

Technical quality folder file consists of the following sections of the CTD:

Table 1. Information required for each section

Section	Information required
3.2. S.3.2 Impurities	• Information on impurities should be provided including classification and identification of impurities, in active drug substances
3.2.P.4.1 Specification of excipients	• Specification of in-house excipients or monograph of compendial excipients
3.2.P.4.6 Novel Excipients	• Novel ExcipientsFor excipient(s) used for the first time in a drug

Section	Information required
	product or by a new route of administration, full details of manufacture, characterization, and controls, with cross references to supporting safety data (nonclinical and/or clinical) should be provided
3.2.P.5.1 Specification of finished pharmaceutical products	• A list of tests, references to analytical procedures and acceptance criteria (which are numerical limits, ranges or other criteria) in a tabulated form. • FPP should conform to the specifications to be considered acceptable for its intended use. • Two separate sets of specifications may be set out: after packaging of the FPP (release) and at the end of shelf-life. • The specifications should be summarized according to the tables including the tests, acceptance criteria and analytical procedures (including types, sources and versions for the methods): **Type**: type of analytical procedure used (e.g. visual, IR, UV, HPLC) Source: reference to the analytical procedure used (e.g. BP, Ph. Eur., Ph.Int, JP, USP, in-house) **Version:** (e.g. code number, version and date)
3.2.P.5.2. Analytical procedures	• Copies of the in-house analytical procedures used should be provided • It isn't necessary to provide copies of officially-recognized Compendial analytical procedures.
3.2.P.5.3 Validation of analytical procedures	• The objective of validation of an analytical procedure is to demonstrate that it is suitable for its intended purpose. • Copies of the validation reports for the in-house analytical procedures used should be provided including: specificity, linearity, accuracy, repeatability, intermediate precision and, for purity: LOD/LOQ. • For officially-recognized Compendial analytical procedures, verification is done.

Section	Information required
	• Verification: is the assessment of whether the Compendial test procedure can be used for its intended purpose, under the actual conditions of use for a specified drug substance and/or drug product matrix. • Revalidation may be necessary if there is a change in the synthesis of the drug substance &/or changes in the composition of the finished product &/or changes in the analytical procedure.
3.2.P.5.4 **Batch analysis**	• Certificate of analysis of batch should be provided • A description of batch (include strength, batch number, date, site of production and results of batch analyses should be provided.
3.2.P.5.5	• Characterization of Impurities
3.2. P.5.6 **Justification of specs**	• A discussion should be provided on the omission or inclusion of particular tests, evolution of tests, analytical procedures and acceptance criteria
3.2.p.6 **Reference standard**	• Information of reference standard used in analysis should be provided

6. History Table

Date	Version Number	Summary of change
2021	**1/2021**	Initial Release
23/05/2024	**2/2024**	• Revision made as needed to align with current Eur. Ph and USP • Revision of limits of antioxidants in finished product specifications (Annex III) • Revision of testing of impurities (organic impurities and residual solvents) to be aligned with ICH Q6A, ICH Q3B(R2) and ICH Q3C(R8) and USP (Annex III) • Revision of method validation and verification requirements (Annex V) to be aligned with ICH Q2(R2) • Revision to formatting and other changes were made in all the document sections to provide policy clarification
13/10/2026	**10/2026**	• Revision of scope and subclause 6.4.1 (Group I) and subclause 6.4.2 (Group II) to align with the regulatory guideline for EDA Chairman Decree No. 450 • Addition of subclause 6.3: Pharmacopoeias that are recognized for CADC • Revision and clarification of Annexes I, II and III • Revision of Annex III: Addition of tests for elemental and mutagenic impurities, to align with ICH Q3D and M7 guidelines • Revision of Annex IV: Addition of tests for probiotics • Revision of technical quality folder requirements (Annex VI)